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Electric device

Abstract

Provided is an electrical device. In a setting unit, a right-room temperature display unit displays a set temperature or the current temperature of a first housing room. A left-room temperature display unit displays a set temperature or the current temperature of a second housing room. A right-room temperature setting button is an operation unit for a user to switch the set temperature of the first housing room. A left-room temperature setting button is an operation unit for a user switch the set temperature of the second housing room. The set temperature difference between the first and second housing rooms is controlled to be within a prescribed value. When the set temperature of one of the first and second housing rooms is changed so that the set temperature exceeds a prescribed value, a microcomputer automatically changes the other set temperature so that the set temperature difference is within a prescribed value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a 371 application of the international PCT application serial no. PCT/JP2022/003216, filed on Jan. 28, 2022, which claims priority benefits of Japan Patent Application No. 2021-018782, filed on Feb. 9, 2021, Japan Patent Application No. 2021-018783, filed on Feb. 9, 2021, Japan Patent Application No. 2021-018784, filed on Feb. 9, 2021, Japan Patent Application No. 2021-018785, filed on Feb. 9, 2021, Japan Patent Application No. 2021-018786, filed on Feb. 9, 2021, Japan Patent Application No. 2021-018787, filed on Feb. 9, 2021, Japan Patent Application No. 2021-018788, filed on Feb. 9, 2021, Japan Patent Application No. 2021-072923, filed on Apr. 22, 2021, Japan Patent Application No. 2021-072924, filed on Apr. 22, 2021, and Japan Patent Application No. 2021-178367, filed on Oct. 29, 2021. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

(2) The present invention relates to an electric device having a cooling or heating function.

BACKGROUND ART

(3) Patent Literature 1 discloses a storage having a cooling/heating function, which includes two accommodation sections partitioned by a partition wall and capable of individually regulating a temperature of each accommodation section. Patent Literature 2 discloses a cold/hot switching type storage that can cool a plurality of accommodation sections with one compressor. Patent Literature

3 discloses a refrigerator with a detachable partition plate.

(4) Patent Literature 4 discloses an electric device capable of cooling the inside of a refrigerator by using a Peltier element, such as a cold-hot storage. Patent Literatures 5 and 6 disclose an electric device capable of cooling the inside of a storage by using a compressor. Patent Literatures 4 and 5 disclose that a battery pack detachable from a main body is used as a drive source and that the battery pack is charged. Patent Literature discloses a fan for cooling a condenser. Patent Literature 6 discloses a condenser (radiator) and an air blower for cooling a control device. Patent Literature 7 discloses an electric device that includes a thermoelectric element that is operated with the power of a battery pack, has a heat or cold insulation space, and can be used outdoors by using the power of the battery pack. A heat or cold insulation storage in Patent Literature 7 includes a DC input part.

CITATION LIST

Patent Literature

(5) [Patent Literature 1]: Japanese Patent Laid-Open No. 2001-194051 [Patent Literature 2]: Japanese Patent Laid-Open No. S58-173367 [Patent Literature 3]: Japanese Patent Laid-Open No. 2013-76494 [Patent Literature 4]: International Publication No. 2018-101144 [Patent Literature 5]: Japanese Patent Laid-Open No. 2017-150700 [Patent Literature 6]: Japanese Patent Laid-Open No. 2015-014434 [Patent Literature 7]: Japanese Patent Laid-Open No. 2018-91501

SUMMARY OF INVENTION

Technical Problem

(6) Patent Literature 1 has a configuration in which the heating/cooling means is disposed on a bottom surface of the accommodation section to cool or heat the bottom surface. Thus, a temperature of the bottom surface side of each accommodation section is easily regulated, but a temperature of the upper side is difficult to regulate. A first objective is to provide an electric device that can individually regulate a temperature in a wide area of each accommodation section and that improves convenience. The present inventor has found a problem that in a case where a plurality of accommodation sections can be cooled by one compressor as in Patent Literature 2, a flow rate of the refrigerant changes when the main body is tilted, and thus a temperature of the accommodation section deviates from a set temperature. In addition, the present inventor has found a problem that in a case where a plurality of accommodation sections can be cooled by one compressor as in Patent Literature 2, a temperature of the accommodation section deviates from a set temperature when two rooms are cooled at the same time due to individual differences between electric devices (dimensional errors in diameters of refrigerant pipes, a difference in a length due to a difference in how the refrigerant pipe is laid, or the like). The present inventor has found the problem that if the set temperatures of a plurality of accommodation sections are set to have temperature differences, temperatures of the respective accommodation sections deviate from the set temperatures. A second objective is to provide an electric device that can accurately control a temperature of an accommodation section even if the electric device has a plurality of accommodation sections.

(7) The present inventor has found a problem that it is difficult to make a temperature of at least one of the accommodation sections reach a set temperature when a difference between set temperatures of the plurality of accommodation sections exceeds a predetermined value. A third objective is to provide an electric device that can reduce the risk that a temperature of an accommodation section cannot reach a set temperature.

(8) The present inventor has found a problem that, in an electric device in which a plurality of accommodation sections is cooled by using a common compressor, if the drive strength of the compressor when cooling only one accommodation section is the same as the driving strength of the compressor when cooling a plurality of accommodation section, a refrigerant may return to the compressor in a liquid state. A fourth objective is to provide an electric device that increases a life of the compressor, that is, reduces a load on the compressor.

(9) The detachable partition plate of the refrigerator in Patent Literature 3 is a single plate, and it is

difficult to carry and store the partition plate. A fifth objective is to provide an electric device with improved convenience in carrying and storing a partition plate.

(10) Patent Literatures 4 and 5 disclose that the battery pack can be charged by a built-in charging circuit, but no consideration is given to cooling the charging circuit. Patent Literature 6 does not disclose how to cool the condenser and the control device. A sixth objective is to provide an electric device capable of efficiently cooling a circuit (circuit board).

(11) The heat or cold insulation storage in Patent Literature 7 has no accommodation function other than the heat insulation or cold insulation space, and is inconvenient in a case where a user wants to carry small items such as a replacement battery pack or a bottle opener. If there is no accommodation function other than the heat or cold insulation space, in a case of employing a configuration having a detachable partition plate, it is inconvenient to carry and place the detached partition plate. A seventh objective is to provide an electric device capable of adding an accommodation function other than a heat or cold insulation space.

(12) In a case where an electric device is used in a vehicle, there is a demand to use an in-vehicle power supply. In the heat or cold insulation storage in Patent Literature 7, it is conceivable to input power from an in-vehicle power supply to the DC input part. However, depending on a state of the in-vehicle power supply, use of the in-vehicle power supply may not be suitable. An eighth objective is to provide an electric device that can appropriately use an in-vehicle power supply and a battery pack.

(13) An objective of the present invention is to provide an electric device that achieves at least the third objective among of the above objectives.

Solution to Problem

(14) One aspect of the present invention is an electric device. The electric device includes a main body having a first accommodation chamber and a second accommodation chamber adjacent to each other and each having a bottom surface and a side surface; a lid capable of being opened and closed with respect to the main body; and a cooling mechanism having a first cooling part that cools the first accommodation chamber and a second cooling part that cools the second accommodation chamber, in which the first cooling part is provided on at least the side surface of the first accommodation chamber, and the second cooling part is provided on at least the side surface of the second accommodation chamber.

Advantageous Effects of Invention

(15) According to the present invention, it is possible to provide an electric device that solves at least the third objective among the above objectives.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of an electric device 1 according to an embodiment of the present invention, viewed from the front upper side, and is a perspective view illustrating a state in which a first lid 6 is open.

(2) FIG. 2 is a perspective view of the electric device 1 viewed from the front upper right, and is a perspective view illustrating a state in which a partition plate 70 is detached from the state in FIG. 1.

(3) FIG. 3 is a perspective view of the electric device 1 viewed from the front upper right, and is a perspective view illustrating a state in which a second lid 7 is open.

(4) FIG. 4 is a perspective view of the electric device 1 viewed from the front upper right, and is a perspective view illustrating a state in which the battery pack 29 detached from the state in FIG. 3.

(5) FIG. 5 is a perspective view of the electric device 1 viewed from the rear upper right.

(6) FIG. 6 is a rear view of the electric device 1.

(7) FIG. 7 is a plan view of the electric device **1** and is a plan view in which the second lid **7** is omitted.

(8) FIG. **8** is an enlarged rear view of the electric device **1** in which a right outer box **13** is omitted.

(9) FIG. **9** is a schematic enlarged view of the inside of the right outer box **13** viewed from above.

(10) FIG. **10** is a drawing for describing attaching a battery box **30** to a main frame **11** in the electric device **1**.

(11) FIG. **11** is a bottom view of the main frame **11** to which the battery box **30** is attached.

(12) FIG. **12** is a perspective view of the electric device **1** viewed from the rear right, and is a perspective view in which the right outer box **13** is omitted.

(13) FIG. **13** is a perspective view of the inside of the electric device **1** viewed from the upper right rear.

(14) FIG. **14** is a front view of the inside of the electric device **1** and is a front view in which a heating mechanism **50** is omitted.

(15) FIG. **15** is a drawing for describing an assembly of a bottom member **15**, a right-side member **16**, a left-side member **17**, and a rail member **18** in the electric device **1**.

(16) (A) of FIG. **16** is a right side view of the inside of the electric device **1** in a case where the electric device **1** is in a horizontal state. (B) of FIG. **16** is a right side view of the inside of the electric device **1** in a case where the electric device **1** is in an inclined state.

(17) FIG. **17** is a simplified block diagram of a mechanical configuration of the electric device **1**.

(18) (A) of FIG. **18** is a right side view of the inside of an electric device according to a modification example in a case where the electric device is in a horizontal state. (B) of FIG. **18** is a right side view of the inside of the electric device of the modification example in a case where the electric device is in an inclined state.

(19) FIG. **19** is a simplified block diagram of a mechanical configuration of the electric device according to the modification example.

(20) FIG. **20** is a front view of the inside of the electric device **1**.

(21) FIG. **21** is a rear view of the inside of the electric device **1**.

(22) FIG. **22** is an enlarged sectional view illustrating an overlapping part of the right-side member **16** (or the left-side member **17**), a first refrigerant pipe **45** (or a second refrigerant pipe **46**), the first heating mechanism **51** (or a second heating part **52**) in the electric device **1**.

(23) FIG. **23** is a perspective view of the left outer box **12** of the electric device **1** and the inside thereof viewed from the upper right rear.

(24) FIG. **24** is a right side view of the same.

(25) (A) of FIG. **25** is a sectional view taken along the line A-A in FIG. **24**. (B) of FIG. **25** is an enlarged view of a B part in (A) of FIG. **25**.

(26) FIG. **26** is a schematic perspective view illustrating a state in which the lower partition plate **72** is attached without attaching an upper partition plate **71** to the bottom member **15**, the right-side member **16**, the left-side member **17**, and the rail member **18** of the electric device **1**, and an accommodation object **79** is accommodated.

(27) (A) of FIG. **27** is a view of the partition plate **70** of the electric device **1** viewed from a surface normal direction. (B) of FIG. **27** is a sectional view taken along line C-C in (A) of FIG. **27**. (C) of FIG. **27** is a perspective view of the partition plate **70**.

(28) (A) of FIG. **28** is a view of the partition plate **70** viewed from the surface normal direction in a state in which the upper partition plate **71** and the lower partition plate **72** are separated. (B) of FIG. **28** is a sectional view taken along line D-D in (A) of FIG. **28**. (C) of FIG. **28** is a perspective view of the partition plate **70** in a state in which the upper partition plate **71** and the lower partition plate **72** are separated.

(29) (A) of FIG. **29** is a diagram of a partition plate **170** having another configuration viewed from the surface normal direction. (B) of FIG. **29** is a sectional view taken along line E-E in (A) of FIG. **29**. (C) of FIG. **29** is a perspective view of the partition plate **170**.

(30) (A) of FIG. 30 is a diagram of the partition plate 170 viewed from the surface normal direction while being folded. (B) of FIG. 30 is a sectional view taken along line F-F in (A) of FIG. 30. (C) of FIG. 30 is a perspective view of the partition plate 170 while being folded.

(31) (A) of FIG. 31 is a diagram of the partition plate 170 viewed from the surface normal direction in a state of being folded. (B) of FIG. 31 is a sectional view taken along line G-G in (A) of FIG. 31. (C) of FIG. 31 is a perspective view illustrating a state in which the partition plate 170 is folded.

(32) (A) of FIG. 32 is an external view of a setting part 60 of the electric device 1. (B) of FIG. 32 is a diagram illustrating a display example of a display part 61 of the setting part 60 in a two-room mode in which temperatures of a first accommodation chamber 9 and a second accommodation chamber 10 are individually controlled. (C) of FIG. 32 is a diagram illustrating a display example of the display part 61 in a large room single mode in which a temperature of only the first accommodation chamber 9 side is controlled. (D) of FIG. 32 is a diagram illustrating a display example of the display part 61 in a small room single mode in which a temperature of only the second accommodation chamber 10 side is controlled. (E) of FIG. 32 is a diagram illustrating a display example of the display part 61 in a one-room mode in which temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 are collectively controlled. (F) of FIG. 32 is a diagram illustrating a display example of the display part 61 in a state in which a set temperature of the first accommodation chamber 9 is 0° C. and a set temperature of the second accommodation chamber 10 is 60° C. in the two-room mode. (G) of FIG. 32 is a diagram illustrating a display example of the display part 61 after the set temperature of the second accommodation chamber 10 is automatically changed to 50° C. in a case where the set temperature of the first accommodation chamber 9 is changed to -10° C. from the state in (F) of FIG. 32.

(33) (A) of FIG. 33 is a table illustrating examples of an operation on the setting part 60, display corresponding to the operation, and a flow of mode switching. (B) of FIG. 33 is a table illustrating examples of an operation on a right room temperature setting button 62 or a left room temperature setting button 63 of the setting part 60 and display corresponding to the operation. (C) of FIG. 33 is a table illustrating a display example of a remaining battery level display part 61a of the display part 61 according to a state of a battery pack 29. (D) of FIG. 33 is a table illustrating a display example of an external power supply connection display part 61b of the display part 61 according to the presence or absence of input of an external DC power supply. (E) of FIG. 33 is a table illustrating a display example of a USB device energization display part 61c of the display part 61 according to the presence or absence of the supply of power to the USB device. (F) of FIG. 33 is a table illustrating a display example of an error display part 61d of the display part 61 when an error occurs.

(34) FIG. 34 is a circuit block diagram of the electric device 1.

(35) FIG. 35 is a circuit block diagram illustrating an internal configuration of a rotation speed setting circuit 84 in FIG. 34.

(36) (A) of FIG. 36 is a time chart illustrating power supply selection, a voltage of a DC power supply 90, and a drive current for a compressor in a case where the DC power supply 90 is a normal in-vehicle power supply. (B) of FIG. 36 is a time chart illustrating power supply selection, a voltage of the DC power supply 90, and a drive current for a compressor 41 in a case where the DC power supply 90 is an in-vehicle power supply in which deterioration has progressed in an operation of the electric device 1 in a comparative example.

(37) FIG. 37 is a time chart illustrating power supply selection, a voltage of the DC power supply 90, a voltage of the battery pack 29a or 29b, a voltage input to a compressor drive circuit 48, and a drive current for the compressor 41 in a case where the DC power supply 90 is an in-vehicle power supply in which deterioration has progressed in an operation of the electric device 1 of the embodiment.

(38) FIG. 38 is a flowchart illustrating a main routine of the electric device 1.

(39) FIG. 39 is a flowchart illustrating a power supply selection routine in FIG. 38.

- (40) FIG. 40 is a flowchart illustrating a DC power state determination routine in FIG. 39.
- (41) FIG. 41 is a flowchart illustrating a charging routine of the electric device 1.
- (42) FIG. 42 is a flowchart illustrating an operation of the electric device 1 in the large room single mode.
- (43) FIG. 43 is a flowchart illustrating an operation of the electric device 1 in the small room single mode.
- (44) FIG. 44 is a flowchart illustrating an operation of the electric device 1 in the one-room mode.
- (45) FIG. 45 is a flowchart illustrating an operation of the electric device 1 in the two-room mode.
- (46) FIG. 46 is a graph illustrating temporal changes in temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 and a rotation speed of the compressor 41 in a case where an operation is performed in the two-room mode when a set temperature of the first accommodation chamber 9 is 10° C. and a set temperature of the second accommodation chamber 10 is -18° C.
- (47) FIG. 47 is a graph illustrating temporal changes in temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 and a rotation speed of the compressor 41 in a case where an operation is performed in the two-room mode when a set temperature of the first accommodation chamber 9 is -18° C. and a set temperature of the second accommodation chamber 10 is 10° C.
- (48) FIG. 48 is a time chart illustrating a voltage input to the compressor 41, a rotation speed of the compressor 41, opening and closing signals for a first regulating valve 47a and a second regulating valve 47b, turning-on and turning-off of a first heating part 51 and a second heating part 52, and temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 in the two-room mode.
- (49) FIG. 49 is a flowchart illustrating an operation in a two-room mode in which an alternating operation mode is added to the two-room mode in FIG. 45.
- (50) FIG. 50 is an explanatory diagram illustrating a specific example of determination 1 (S521) in FIG. 49.
- (51) FIG. 51 is an explanatory diagram illustrating a specific example of determination 2 (S522) in FIG. 49.
- (52) FIG. 52 is an explanatory diagram illustrating a specific example of determination 3 (S80) in FIG. 53.
- (53) FIG. 53 is a flowchart illustrating a first pattern of the alternating operation mode (S523) in FIG. 49.
- (54) FIG. 54 is a flowchart illustrating a second pattern of the alternating operation mode (S523) in FIG. 49.
- (55) FIG. 55 is a graph (part 1) illustrating temporal changes in temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 and a rotation speed of the compressor 41 in a case where set temperatures of both the first accommodation chamber 9 and the second accommodation chamber 10 are -18° C., and an operation associated with transition from a simultaneous operation mode to the alternating operation mode is performed in the two-room mode.
- (56) FIG. 56 is a graph (part 2) illustrating temporal changes in temperature of the first accommodation chamber 9 and the second accommodation chamber 10 and a rotation speed of the compressor 41 in a case where set temperatures of both the first accommodation chamber 9 and the second accommodation chamber 10 are -18° C., and an operation associated with transition from the simultaneous operation mode to the alternating operation mode is performed in the two-room mode.
- (57) FIG. 57 is a graph illustrating temporal changes in temperature of the first accommodation chamber 9 and the second accommodation chamber 10 and a rotation speed of the compressor 41 in a case where a set temperature of the first accommodation chamber 9 is -18° C., a set temperature

of the second accommodation chamber **10** is -5°C ., and an operation associated with transition from the simultaneous operation mode to the alternating operation mode is performed in the two-room mode.

(58) FIG. **58** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is 60°C ., a set temperature of the second accommodation chamber **10** is 30°C ., and cooling of the second accommodation chamber **10** using the compressor **41** is stopped by controlling a temperature of the second accommodation chamber **10** in the two-room mode.

(59) FIG. **59** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is 60°C ., a set temperature of the second accommodation chamber **10** is 30°C ., and cooling of the second accommodation chamber **10** using the compressor **41** is stopped by controlling a time from the start of the cooling in the two-room mode.

(60) FIG. **60** is a flowchart illustrating an operation in a two-room mode in which time control is added to the two-room mode in FIG. **45**.

(61) (A) of FIG. **61** is a waveform diagram of a drive current for the compressor **41** before and after an operation is resumed in a case where an operation of the compressor **41** is performed, stopped for one minute, and then resumed. (B) of FIG. **61** is a waveform diagram of a drive current for the compressor **41** before and after an operation is resumed in a case where an operation of the compressor **41** is performed, stopped for two minutes, and then resumed.

(62) FIG. **62** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is 60°C ., a set temperature of the second accommodation chamber **10** is 30°C ., and cooling of the second accommodation chamber **10** using the compressor **41** is performed by repeatedly performing an operation for one minute and stoppage for two minutes in the two-room mode.

(63) FIG. **63** is a perspective view of main parts of the electric device **1** with a storage unit (additional part) **200** attached to the right side surface.

(64) (A) of FIG. **64** is a front view of the same. (B) of FIG. **64** is a right side view of the same.

(65) (A) of FIG. **65** is a plan view of the same. (B) of FIG. **65** is a bottom view of the same.

(66) (A) of FIG. **66** is a front sectional view of the main parts (sectional view taken along the line H-H in (B) of FIG. **66**). (B) of FIG. **66** is a view in which a bag **201** is removed in (B) of FIG. **64**.

(67) (A) of FIG. **67** is a perspective view of the storage unit **200** in a state in which the bag **201** is open. (B) of FIG. **67** is a perspective view of the storage unit **200** in a state in which the bag **201** is closed. (C) of FIG. **67** is a perspective view of the storage unit **200** viewed from the back side. (D) of FIG. **67** is a perspective view of a storage unit in which the bag **201** is replaced with a bag **201A**.

(68) FIG. **68** is six views of the storage unit **200**.

(69) FIG. **69** is six views of a holder part of the storage unit **200**.

(70) FIG. **70** is six views of the bag **201** of the storage unit **200**.

(71) FIG. **71** is a perspective view of the electric device **1** to which the storage unit **200**, accessory cases **231** to **233**, and an S-shaped hook **237** are attached.

(72) FIG. **72** is a perspective view of the electric device **1** in which the accessory case **231** is attached in place of the storage unit **200** in FIG. **63**.

(73) FIG. **73** is an upper perspective view of main parts of the electric device **1** in which a storage unit (additional part) **250** is attached to a movable handle **20**.

(74) FIG. **74** is a lower perspective view of the same.

(75) FIG. **75** is a perspective view of an electric device **1A** having a storage unit (additional part) **300** attached to a front surface thereof according to another embodiment of the present invention.

(76) FIG. 76 is a view in which a bag 301 is removed in FIG. 75.

(77) FIG. 77 is a front perspective view of the electric device 1A in which the storage unit 200 is attached to the right surface, the storage unit 300 is attached to the front surface, and the storage unit (additional part) 400 is attached to the rear surface.

(78) FIG. 78 is a rear perspective view of the same.

(79) FIG. 79 is a front perspective view of the electric device 1A in which a storage unit (additional part) 500 is attached to an upper surface.

(80) FIG. 80 is a rear perspective view of the same.

(81) FIG. 81 is a view in which a bag 501 is removed in FIG. 79.

(82) FIG. 82 is a perspective view of an electric device 1B according to still another embodiment of the present invention.

(83) FIG. 83 is a perspective view of the inside of the electric device 1 in a state in which the accommodation chamber 8 is divided into three by using the partition plates 70a and 70b (three-room mode), viewed from the front upper right.

DESCRIPTION OF EMBODIMENTS

(84) Hereinafter, the same or equivalent constituents, members, and the like illustrated in the respective drawings are denoted by the same reference numerals, and redundant description will be omitted as appropriate. The embodiments are illustrative rather than limiting of the invention. All features and combinations thereof described in the embodiments are not necessarily essential to the invention.

(85) The present embodiment relates to an electric device 1. The electric device 1 is a portable cold-hot storage with cooling and heating functions. With reference to FIG. 1 and the like, front and rear, up and down, and left and right directions in the electric device 1 that are orthogonal to each other are defined. The front-rear direction is a depth direction (lateral direction) in the electric device 1, the left-right direction is a width direction (longitudinal direction) in the electric device 1, and the up-down direction is a height direction in the electric device 1.

(86) The electric device 1 has a main body 2. The main body 2 has a first body part 3 and a second body part 4 different from the first body part 3. The first body part 3 and the second body part 4 are arranged in the left-right direction. The second body part 4 is located outside and to the right of the first body part 3.

(87) The first body part 3 has a left outer box 12. The left outer box 12 is, for example, a resin molded body having an open top and a substantially rectangular parallelepiped shape, and constitutes the exterior of the first body part 3. The second body part 4 has a right outer box 13. The right outer box 13 is, for example, a resin molded body having open top and left side and a substantially rectangular parallelepiped shape, and constitutes the exterior of the second body part 4. The right outer box 13 is fixed and integrated with a right side surface of the left outer box 12 by screws or the like.

(88) The main body 2 has a main frame 11. The main frame 11 is, for example, a resin molded body. The main frame 11 is a frame extending over upper parts of the first body part 3 and the second body part 4, and has openings 11c and 11d respectively corresponding to the first body part 3 and the second body part 4 as illustrated in FIG. 10.

(89) The electric device 1 has a lid 5. The lid 5 is provided on the upper part of the main body 2 and can be opened and closed with respect to the main body 2. The lid 5 has a first lid 6 that opens and closes the first body part 3 and a second lid 7 that opens and closes the second body part 4 (opens and closes a battery pack accommodation chamber 22).

(90) The first lid 6 is rotatably connected to a rear end of the main frame 11 by a first hinge mechanism 25, as illustrated in FIG. 5 and the like. The first lid 6 has a grip part 6a. A user releases locking of the first lid 6 to the main frame 11 by rotating forward the grip part 6a of the first lid 6 in a closed state illustrated in FIG. 3, and opens the first lid 6 while holding the grip part 6a as illustrated in FIG. 1.

(91) The second lid 7 is rotatably connected to a rear end of the main frame 11 by a second hinge mechanism 26 as illustrated in FIG. 5 and the like. The second lid 7 has a grip part 7a. A user releases locking of the second lid 7 by rotating upward the handle 7a of the second lid 7 in a closed state illustrated in FIG. 1, and opens the second lid 7 while holding the handle 7a as illustrated in FIG. 3.

(92) The electric device 1 includes a pair of left and right grip parts 21 and a plurality of (for example, four) legs 35 that serve as grounding parts on the bottom. A user can move the electric device 1 by holding the left and right grip parts 21 and lifting the electric device 1.

(93) The electric device 1 has casters 19 and a movable handle (carry handle) 20. The casters 19 are provided at the front and rear of the lower right part of the main body 2, respectively. A rotation axis direction of the caster 19 is parallel to the left-right direction. The movable handle 20 is rotatably provided on the left side surface of the main body 2. A rotation axis direction of the movable handle 20 is parallel to the left-right direction. A user can move the electric device 1 by using the casters 19 by rotating the movable handle upward, holding the movable handle 20 and lifting the left part of the main body 2 from the ground.

(94) The electric device 1 includes a USB terminal 27 and a power input terminal 28 on the second body part 4. The electric device 1 can supply charging power to a device connected to the USB terminal 27. The electric device 1 can receive DC power from the outside via the power input terminal 28. The electric device 1 is operated with the DC power or the power of a battery pack 29.

(95) The first body part 3 has an accommodation chamber (accommodation section) 8. The accommodation chamber 8 has a first accommodation chamber 9 which is a large room and a right room, and a second accommodation chamber 10 which is a small room and a left room. The first accommodation chamber (first accommodation section) 9 and the second accommodation chamber (second accommodation section) 10 are adjacent to each other and separated (partitioned) by a detachable partition plate 70.

(96) The first body part 3 has a bottom member 15, a right-side member 16, a left-side member 17, and a rail member 18 illustrated in FIG. 15 and the like, which are constituents of the accommodation chamber 8. The bottom member 15 is, for example, a resin molded body, and constitutes the bottom surface of the accommodation chamber 8. The right-side member 16 and the left-side member 17 are each made of metal (metal plate) such as aluminum and are U-shaped when viewed from above. The right-side member 16 constitutes the side surface of the first accommodation chamber 9. The left-side member 17 constitutes the side surface of the second accommodation chamber 10.

(97) A first side surface 9a of the first accommodation chamber 9 is a right side surface of the first accommodation chamber 9 and a side surface that is the farthest from the second accommodation chamber 10. A second side surface 9b of the first accommodation chamber 9 is a front surface and a rear surface of the first accommodation chamber 9 and is a side surface connected to the first side surface 9a. A first side 10a of the second accommodation chamber 10 is a left side surface of the second accommodation chamber and a side surface that is the farthest from the first accommodation chamber 9. The second side surface 10b of the second accommodation chamber 10 is a front surface and a rear surface of the second accommodation chamber 10, and is a side surface connected to the first side surface 10a.

(98) As illustrated in FIG. 15 and (B) of FIG. 25, the right-side member 16 has outwardly bent part 16a at each of a pair of left ends. The left-side member 17 has an outwardly bent part 17a at each of a pair of right ends. The outwardly bent parts 16a and 17a each extend in the up-down direction.

(99) The pair of rail members 18 has groove parts 18c and 18d into which the outwardly bent parts 16a and 17a are fitted (engaged), respectively. The rail members 18 are fixed and integrated with the right-side member 16 and the left-side member 17 (to the outwardly bent parts 16a, 17a) by screws or the like in a state in which the outwardly bent parts 16a and 17a are fitted in the groove parts 18c and 18d. That is, the rail members 18 also have a function of connecting the right-side

member **16** and the left-side member **17**. The rail member **18** has a notch **18a** for screwing to the main frame **11**. The notch **18a** communicates with a boss **11b** (FIG. **11**) of the main frame **11**. The rail member **18** is fixed to the main frame **11** by a screw (not illustrated) screwed into the boss **11b** through the notch **18a**.

(100) The rail member **18** has a recessed part (recessed groove) **18b** extending in the up-down direction. The recessed part **18b** constitutes a recessed part on the inner surface of the accommodation chamber **8** and serves as a guide in a case where the partition plate **70** is attached and detached. The front rail member **18** does not protrude rearward from the inside of the front surfaces of the right-side member **16** and the left-side member **17**, that is, the inside of the front surface of the accommodation chamber **8**. The rear rail member **18** does not protrude forward from the inside of the rear surfaces of the right-side member **16** and the left-side member **17**, that is, the inside of the rear surface of the accommodation chamber **8**.

(101) The right-side member **16**, the left-side member **17**, and the rail member **18** that are combined with each other are fitted into the bottom member **15** from above to form an inner box of the first body part **3**. A heat insulating material (not illustrated) is filled between the inner box and the left outer box **12**. This heat insulating material hardens after filling the heat insulating material, and thus also serves to fix the inner box to the left outer box **12** to further fix the main frame **11**.

(102) As illustrated in FIG. **15**, the electric device **1** includes a first thermistor **55** as a first temperature sensor and a first holder **57** holding the first thermistor, and a second thermistor **56** as a second temperature sensor and a second holder **58** holding the second thermistor. The first holder **57** is fixed to a substantially central part of the right side surface of the right-side member **16** by screws or the like. The first thermistor **55** detects the temperature inside the right-side member **16**, that is, inside the first accommodation chamber **9**. The second holder **57** is fixed to a substantially central part of the left side surface of the left-side member **17** by screws or the like. The second thermistor **56** detects the temperature inside the left-side member **17**, that is, inside the second accommodation chamber **10**. The electric device **1** may include a third temperature sensor that detects the temperature outside the first accommodation chamber **9** and the second accommodation chamber **10**, that is, the outside air temperature.

(103) The second body part **4** has a battery pack accommodation chamber **22** as illustrated in FIG. **4**. The battery pack accommodation chamber **22** has two battery pack attachment parts **22a** (battery pack connection parts). As illustrated in FIG. **3**, the battery pack **29** can be detachably attached (connected) to each battery pack attachment part **22a**. The battery pack **29** is, for example, a battery pack for power tools with a rated output voltage of 18 V. The battery pack accommodation chamber **22** is a space formed by a battery box **30** and a battery terminal **31** illustrated in FIG. **10** and the like.

(104) The battery box **30** is, for example, a resin molded body. The battery terminal **31** is, for example, a resin molded body that holds a connection terminal with the battery pack **29**. As illustrated in FIG. **10**, the battery box **30** is inserted into the opening **11d** of the main frame **11** from above and fixed to the main frame **11** with screws. As illustrated in FIG. **11**, the battery box **30** has bosses **30a** for screwing to the main frame **11**. A bottom surface of the battery box **30** is provided with a drain hole **30b**.

(105) As illustrated in FIG. **14**, the electric device **1** has a cooling mechanism **40**. The cooling mechanism **40** can cool the accommodation chamber **8** with DC power input from the outside via the power input terminal **28** or power from the battery pack **29**. The cooling mechanism **40** has a compressor **41** as a cooler, a condenser **42**, a refrigerant pipe **44** and a regulating valve **47**. The cooling mechanism **40** is mostly provided in the second body part **4**, that is, except for a portion of the refrigerant pipe **44** that extends into the first body part **3**. The condenser **42** is provided close to the inner wall surface of the second body part **4** opposite to the first body part **3**, that is, the inner surface of the right side surface of the right outer box **13**, and has a space between the condenser **42** and right side surface of the left outer box **12**.

(106) The compressor **41** is an output part of the electric device **1** and has a motor, compresses a refrigerant, and discharges the refrigerant as a high-temperature and high-pressure gas. The condenser **42** releases the heat of the refrigerant discharged from the compressor **41** and discharges the refrigerant as a liquid. The refrigerant pipe **44** constitutes a path along which the refrigerant discharged from the condenser **42** passes around the accommodation chamber **8** and returns to the compressor **41**. The refrigerant draws heat from the accommodation chamber **8** while passing around the accommodation chamber **8** and evaporates into a gas. The regulating valve **47** is provided in a portion of the refrigerant pipe **44** where the refrigerant returns from the surroundings of the accommodation chamber **8** to the compressor **41**, that is, in a portion where the refrigerant flows in a gaseous state.

(107) The refrigerant pipe **44** has a first refrigerant pipe **45** as a first cooling part and a second refrigerant pipe **46** as a second cooling part. The single refrigerant pipe **44** extends into the first body part **3**, and, as illustrated in FIG. **13**, divided into two pipes, that is, the first refrigerant pipe **45** and the second refrigerant pipe **46** at a branch part **44a** near the rear surface of the right-side member **16**. That is, the single refrigerant pipe **44** extends to the branch part **44a** near the boundary (rail **18**) between the first accommodation chamber **9** and the second accommodation chamber **10**. As a result, the surface area of the refrigerant pipe in contact with the outside air is smaller than that in a case where the refrigerant pipe is branched near the condenser, and the unnecessary temperature rise of the refrigerant can be suppressed, so that the accommodation chamber can be cooled more efficiently. The refrigerant pipe forming a path from the regulating valve **47** to a suction part of the compressor **41** is an example of a third cooling part, and the refrigerant pipe forming the discharge part of the compressor **41** to the branch part **44a** is an example of a fourth cooling part.

(108) The first refrigerant pipe **45** is provided on at least the side surface of the first accommodation chamber **9** and cools the first accommodation chamber **9**. The first refrigerant pipes **45** are provided at least on the first side surface **9a**, preferably on the first side surface **9a** and the second side surface **9b**. The second refrigerant pipe **46** is provided on at least the side surface of the second accommodation chamber **10** and cools the second accommodation chamber **10**. A second refrigerant pipe **46** is provided at least on the first side surface **10a**, preferably on the first side surface **10a** and the second side surface **10b**. The first refrigerant pipe **45** and the second refrigerant pipe **46** are independent of each other. That is, the first refrigerant pipe **45** is provided to mainly cool the first accommodation chamber **9**, and the second refrigerant pipe **46** is provided to mainly cool the second accommodation chamber **10**.

(109) The first refrigerant pipe **45** forms a path in which reciprocation of extending along the right-side member **16** from the left end of the back surface of the right-side member **16** to the center of the right side surface of the right-side member **16** in the front-rear direction, and then folding back to the left end of the back surface of the right-side member **16** is repeated three times, then reciprocation of extending from the left end of the back surface of the right-side member **16** to the left end of the front surface of the right-side member **16**, further extending from the left end of the front surface of the right-side member **16** to the center of the right side surface of the right-side member **16** in the front-rear direction, and then folding back to the left end of the front surface of the right-side member **16** is repeated three times, and then the first refrigerant pipe **45** reaches the regulating valve **47**. A gas reservoir **45a** is provided immediately before the regulating valve **47** to temporarily store the expanded refrigerant before returning to the compressor.

(110) The second refrigerant pipe **46** forms a path in which reciprocation of extending along the left-side member **17** from the right end of the back surface of the left-side member **17** to the center of the left side surface of the left-side member **17** in the front-rear direction, and then folding back to the right end of the back surface of the left-side member **17** is repeated three times, then reciprocation of extending from the right end of the back surface of the left-side member **17** to the right end of the front surface of the left-side member **17**, further extending from the right end of the

front surface of the left-side member **17** to the center of the left side surface of the left-side member **17** in the front-rear direction, and folding back to the right end of the front surface of the left-side member **17**, and then the second refrigerant pipe **46** reaches the regulating valve **47**. A gas reservoir **46a** is provided immediately before the regulating valve **47** to temporarily store the expanded refrigerant before returning to the compressor. Here, the gas reservoir **45a** and the gas reservoir **46a** are examples of reservoir parts.

(111) As illustrated in FIG. **15**, at least a part of the first refrigerant pipe **45** is integrally provided with the right-side member **16**, and at least a part of the second refrigerant pipe **46** is integrally provided with the left-side member **17**. In other words, the right-side member **16** and the left-side member **17** are integrated refrigerant pipes.

(112) The regulating valve **47** is connected to the first refrigerant pipe **45** and the second refrigerant pipe **46** and can individually regulate (open/close) the flow of the refrigerant in the first refrigerant pipe **45** and the second refrigerant pipe **46**. The regulating valve **47** includes a first regulating valve **47a** provided in the first refrigerant pipe **45** to regulate the flow of the refrigerant in the first refrigerant pipe **45**, and a second regulating valve **47b** provided in the second refrigerant pipe **46** to regulate the flow of the refrigerant in the second refrigerant pipe **46**. For example, by opening only the first regulating valve **47a**, the refrigerant flows through the first refrigerant pipe **45** as a path, so that the first accommodation chamber **9** can be cooled. This is because a difference ($P1 < P2$) occurs between the internal pressure ($P1$) of the refrigerant pipe **45** and the internal pressure ($P2$) of the refrigerant pipe **46** at the branch part **44a**. In a case where it is desired to cool both the first accommodation chamber **9** and the second accommodation chamber **10**, both the first regulating valve **47a** and the second regulating valve **47b** are opened to allow the refrigerant to flow through the first refrigerant pipe **45** and the second refrigerant pipe **46** as a path, and thus both the first accommodation chamber **9** and the second accommodation chamber **10** can be cooled. Here, the regulating valve **47** has the functions of the first regulating valve **47a** and the second regulating valve **47b** in a single form, but the first regulating valve **47a** and the second regulating valve **47b** may be separate from each other. The first regulating valve **47a** and the second regulating valve **47b** are, for example, solenoid valves.

(113) Providing the first regulating valve **47a** and the second regulating valve **47b** in the portions of the first refrigerant pipe **45** and the second refrigerant pipe **46**, respectively, through which the refrigerant flows in a gaseous state, contributes to suppressing imbalance in an amount of refrigerant flowing through the first refrigerant pipe **45** and the second refrigerant pipe **46** in a case where the electric device **1** is in an inclined state. This is because the refrigerant is light in a gaseous state, and thus even if there is a height difference between the positions of the first regulating valve **47a** and the second regulating valve **47b**, the uneven flow of the refrigerant is less likely to occur. (A) and (B) of FIG. **16** visually illustrate that the flow of the refrigerant is less likely to become uneven regardless of whether the electric device **1** is in a horizontal state or an inclined state. (A) and (B) of FIG. **16** illustrate the first regulating valve **47a** and the second regulating valve **47b** as separate bodies.

(114) On the other hand, unlike the present embodiment, as illustrated in (A) and (B) of FIG. **18**, there may be a configuration in which a first regulating valve **147a** and a second regulating valve **147b** are respectively provided in portions of the first refrigerant pipe **45** and the second refrigerant pipe **46** through which the refrigerant flows in a liquid state. In the configuration of the electric device according to this modification example, the flow is favorable in a horizontal state illustrated in (A) of FIG. **18**, but in a case of an inclined state, the flow of the refrigerant is uneven as illustrated in (B) of FIG. **18**. This is because the refrigerant is heavy in a liquid state, and thus a height difference between the positions of the first regulating valve **147a** and the second regulating valve **147b** influences the flow of the refrigerant. Thus, in a case where the electric device **1** is used in an inclined state, the configuration of the electric device **1** of the present embodiment is preferable.

(115) In the electric device **1**, as illustrated in FIG. **17**, the gaseous refrigerant discharged from the compressor **41** passes through the condenser **42** and the capillary tube **43**, flows into the refrigerant pipe **44** in a liquid state, branches into the first refrigerant pipe **45** and the second refrigerant pipe **46**, absorbs heat from the first accommodation chamber **9** and the second accommodation chamber **10** to be vaporized in a gaseous state, and passes through the first regulating valve **47a** and the second regulating valve **47b** to return to the compressor **41**. On the other hand, in the electric device of the modification example illustrated in (A) and (B) of FIG. **18**, as illustrated in FIG. **19**, the gaseous refrigerant discharged from the compressor **41** passes through the first regulating valve **147a** and the second regulating valve **147b** in a liquid state via the condenser **42** and the capillary tube **43**, flows into the refrigerant pipe **44**, branches into the first refrigerant pipe and the second refrigerant pipe **46**, absorbs heat from the first accommodation chamber **9** and the second accommodation chamber **10** to be vaporized, and returns to the compressor **41** in a gaseous state.

(116) The electric device **1** has a fan **49** as illustrated in FIGS. **8** and **12** and the like. The fan **49** is provided inside the right outer box **13**. That is, the fan **49** is accommodated in the second body part **4**. The fan **49** generates fan air for cooling the cooling mechanism (especially the condenser **42**) and a control circuit board **80** that will be described later. An intake port **23** for taking in fan air is provided on the front surface of the right outer box **13**. An exhaust port **24** for discharging the fan air is provided on the back surface of the right outer box **13**. The fan air flows from the intake port **23** toward the exhaust port **24** in the front-rear direction. The front-to-rear relationship of intake and exhaust may be reversed.

(117) The electric device **1** includes the control circuit board **80** as illustrated in FIGS. **8** and **12** and the like. The control circuit board **80** is fixed to the bosses **12a** provided on the outer surface of the right side surface of the left outer box **12** (the boundary wall surface between the first body part **3** and the second body part **4**) by screws, and is located in the right outer box **13** in a state substantially perpendicular to the left-right direction. That is, the control circuit board **80** is accommodated in the second body part **4**. The control circuit board **80** is equipped with an internal circuit surrounded by a dashed line in FIG. **34**, and has a function of controlling the cooling mechanism **40** and a function of controlling charging of the battery pack **29**. The charging here is to charge the battery pack **29** attached to the battery pack attachment part **22a** with DC power from the outside via the power input terminal **28**.

(118) The electric device **1** includes a branch part **32** as illustrated in FIGS. **8**, **9**, and **12**, and the like. The branch part **32** divides the fan air into a first air passage **33** for cooling the cooling mechanism **40** (especially the condenser **42**) and a second air passage **34**, separate from the first air passage **33**, for cooling the control circuit board **80**. The branch part **32** is a plate-like part extending downward from the bottom surface of battery pack accommodation chamber **22** and substantially perpendicular to the left-right direction. The branch part **32** is preferably integral with the battery box **30**.

(119) The electric device **1** includes a heating mechanism **50** capable of heating the first accommodation chamber **9** and the second accommodation chamber **10**, as illustrated in FIGS. **20** to **22**. The heating mechanism **50** has a first heating part **51** and a second heating part **52**. The first heating part **51** is provided on the outer surface of the right-side member **16** (the outer surface of the first accommodation chamber **9**) and can heat the first accommodation chamber **9**. The second heating part **52** is provided on the outer surface of the left-side member **17** (the outer surface of the second accommodation chamber **10**) and can heat the second accommodation chamber **10**. The first heating part **51** and the second heating part **52** are wire heaters, for example, and are provided to cover the first refrigerant pipe **45** and the second refrigerant pipe **46**, respectively. If a single heating part or a single refrigerant pipe is disposed to cover the entire accommodation chamber and a rail member is provided, the heating part or the refrigerant pipe is required to be disposed to avoid the rail member **18**. As a result, a distance between the heating part or the refrigerant pipe and the accommodation chamber **8** increases, and thus the heating performance or the cooling

performance may deteriorate. However, in the present embodiment, the first heating part **51** and the second heating part **52** and the first refrigerant pipe **45** and the second refrigerant pipe **46** are each divided, and the rail member **18** is disposed in the divided part. Therefore, it is not necessary to increase a distance between the first heating part **51** and the second heating part **52** and the first refrigerant pipe **45** and the second refrigerant pipe **46** from the accommodation chamber **8**, and thus it is possible to provide an electric device in which the partition plate **70** can be attached and detached while maintaining the heating performance or the cooling performance.

(120) The electric device **1** has a partition plate **70**. As illustrated in FIG. **1**, the partition plate **70** divides the accommodation chamber **8** into the first accommodation chamber **9** and the second accommodation chamber **10** in a state of being attached to the main body **2**. A user can attach and detach the partition plate **70** to and from the accommodation chamber **8** along the recessed part **18b** ((B) of FIG. **25**) of the rail member **18**.

(121) The partition plate **70** can be divided into an upper partition plate **71** and a lower partition plate **72** as illustrated in (A) to (C) of FIG. **27** and (A) to (C) of FIG. **28**. As illustrated in (B) of FIG. **27** and (B) of FIG. **28**, the upper partition plate **71** and the lower partition plate **72** each have a heat insulating material **74** in their internal spaces (each internal space is filled with the heat insulating material **74**). A mating part **73** between the upper partition plate **71** and the lower partition plate **72** has a spigot structure. That is, the upper partition plate **71** and the lower partition plate **72** partially overlap each other in the up-down direction, and the overlapping parts approach or contact each other in the surface normal direction. The heat insulating material **74** extends inside the spigot structure. The partition plate **70** detached from the accommodation chamber **8** can be stored in a storage part (not illustrated) detachably provided in the main body **2**. The storage part can accommodate not only the partition plate **70** but also a battery, a smart phone, and the like, and has a size that can accommodate at least the partition plate **70** in a divided state.

(122) The upper partition plate **71** and the lower partition plate **72** function as a heat insulating material (heat insulating wall) between the first accommodation chamber **9** and the second accommodation chamber **10** in a state in which the upper partition plate **71** and the lower partition plate **72** are combined with each other and attached to the main body **2** (accommodation chamber **8**). As illustrated simply in FIG. **26**, the lower partition plate **72** functions as a scattering prevention wall for an accommodation object **79** accommodated in the accommodation chamber **8** in a state in which the lower partition plate **72** is attached to the accommodation chamber **8** alone without being combined with the upper partition plate **71**.

(123) (A) to (C) of FIG. **29** and (A) to (C) of FIG. **30** illustrate a configuration of a partition plate **170** that can be attached to the accommodation chamber **8** in place of the partition plate **70**. The partition plate **170** has an upper partition plate **171** and a lower partition plate **172**. The upper partition plate **171** and the lower partition plate **172** are connected to each other via a hinge mechanism **175** and configured to be foldable. Each internal space of the upper partition plate **171** and the lower partition plate **172** is filled with a heat insulating material **174**. A mating part **173** between the upper partition plate **171** and the lower partition plate **172** has a spigot structure. The heat insulating material **174** extends inside the spigot structure.

(124) The electric device **1** includes a setting part **60**. The setting part **60** is provided at the upper right front end of the main body **2** and faces forward and upward. A user can individually set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** by using the setting part **60**. For example, one of the first accommodation chamber **9** and the second accommodation chamber **10** may be set to be frozen and the other may be set to be refrigerated or set to a normal temperature, one may be set to be frozen or refrigerated and the other may be set to be heated or set to a normal temperature, or one may be set to be strongly heated, and the other may be set to be weakly heated or set to a room temperature. The partition plate **70** (or the partition plate **170**) suppresses heat transfer between the first accommodation chamber **9** and the second accommodation chamber **10** in a case where the set temperatures of the first accommodation

chamber **9** and the second accommodation chamber **10** are different. In other words, by attaching the partition plate **70** (or the partition plate **170**), it is possible to control the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** to be different from each other.

(125) As illustrated in (A) of FIG. **32**, the setting part **60** includes a display part **61**, a right room temperature setting button **62**, a left room temperature setting button **63**, a mode switching button **64** (room switching button), a power button **65**, a USB device energization switching button **66**, and an execution button **67**. The display part **61** includes a remaining battery level display part **61a**, an external power supply connection display part **61b**, a USB device energization display part **61c**, an error display part **61d**, a right room temperature display part **61e**, and a left room temperature display part **61f**. Display on the display part **61** is controlled by a microcomputer **81** illustrated in FIG. **34**. An operation on each button is transmitted to the microcomputer **81**.

(126) As illustrated in (C) of FIG. **33**, the battery state display part **61a** displays a state of the battery pack **29** in the battery pack attachment part **22a**. While the battery pack **29** is being charged, the battery state display part **61a** repeats display of a remaining capacity in increments of 1, 2, 3, and 4. In a case where the charging of the battery pack **29** is completed, four remaining capacity indicators are lighted in the battery state display part **61a**. In a case where the electric device **1** is operated with the power of the battery pack **29**, in the battery state display part **61a**, the remaining capacity display of the battery pack **29** in use blinks, and the remaining capacity display of the battery pack **29** not in use is lighted. In a case where the remaining capacity of the battery pack **29** is low (below a dischargeable threshold value), only the remaining capacity display frame is lighted in the battery state display part **61a**.

(127) As illustrated in (D) of FIG. **33**, the external power supply connection display part **61b** is lighted in a case where DC power is input from the outside via the power input terminal **28**, and lighted off in other cases. As illustrated in (E) of FIG. **33**, the USB device energization display part **61c** is lighted in a case where charging power is being supplied to a device connected to the USB terminal **27**, and lighted off in other cases. As illustrated in (F) of FIG. **33**, the error display part **61d** blinks and displays an error code in a case where an error occurs, and is lighted off in other cases. The error is, for example, an abnormal temperature of the compressor **41**, an abnormal overcurrent, an abnormal input power supply (low voltage), or an abnormality in which a set temperature deviates from a temperature settable range.

(128) The right room temperature display part **61e** displays a set temperature or the current temperature of the first accommodation chamber **9**. The left room temperature display part **61f** displays a set temperature or the current temperature of the second accommodation chamber **10**. For example, by displaying the set temperature in a blinking manner and displaying the current temperature in a lighting manner, the set temperature and the current temperature can be displayed separately on the same display part.

(129) The right room temperature setting button **62** is an operation part for a user to switch a set temperature of the first accommodation chamber **9**. The left room temperature setting button **63** is an operation part for a user to switch a set temperature of the second accommodation chamber **10**. As illustrated in (B) of FIG. **33**, pressing the upper parts of the right room temperature setting button **62** and the left room temperature setting button **63** raises the set temperature by 5° C., and pressing the lower parts lowers the set temperature by 5° C. For example, an initial set temperature is 10° C., the maximum set temperature is 60° C., and the minimum set temperature is -18° C.

(130) The mode switching button **64** is an operation part for a user to switch operation modes of the electric device **1**. The operation modes include a two-room mode in which temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** are individually controlled, a large room single mode (right room single mode) in which a temperature of only the first accommodation chamber **9** side is controlled, a small room single mode (left room single mode) in which a temperature of only the second accommodation chamber **10** side is controlled, and a one-

room mode in which temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** are collectively controlled. For example, the initial operation mode is the two-room mode, and each time the mode switching button **64** is pressed, the mode changes among the large room single mode, the small room single mode, and the one-room mode.

(131) As illustrated in (B) of FIG. **32**, in the two-room mode, the right room temperature display part **61e** and the left room temperature display part **61f** respectively display set temperatures or current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**, respectively. As illustrated in (C) of FIG. **32**, in the large room single mode, the right room temperature display part **61e** displays the set temperature or the current temperature of the first accommodation chamber **9**, while the left room temperature display part **61f** is lighted off. As illustrated in (D) of FIG. **32**, in the small room single mode, the right room temperature display part **61e** is lighted off, while the left room temperature display part **61f** displays a set temperature or the current temperature of the second accommodation chamber **10**. As illustrated in (E) of FIG. **32**, in the one-room mode, the right room temperature display part **61e** displays the set temperatures or the current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**, and the left room temperature display part **61f** is lighted off. The large room single mode and the one-room mode are distinguished by whether or not a partition line between the right room temperature display part **61e** and the left room temperature display part **61f** is lighted, as illustrated in (C) and (E) of FIG. **32** (lighting: large room single mode, lighting-off: one-room mode).

(132) The power button **65** is an operation part for a user to switch between starting and stopping of the electric device **1**. The USB device energization switching button **66** is an operation part for a user to switch whether or not to supply charging power to a device connected to the USB terminal **27**. The execution button **67** is a button for a user to determine the current set temperature and start an operation at the set temperature. The execution button **67** may be omitted, and an operation at a set temperature may be started in conjunction with a user's operation on the operation part.

(133) In the electric device **1**, a set temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** is controlled to be within a predetermined value. This takes into consideration the limit of the maximum temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** that can be realized by the heat insulating effect of the partition plate **70** (or the partition plate **170**). (F) and (G) of FIG. **32** illustrate that a set temperature of the second accommodation chamber **10** is automatically changed to 50° C. when control is performed such that a set temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** is within 60° C., in a case where a set temperature of the first accommodation chamber **9** is changed to -10° C. from a state in which a set temperature of the first accommodation chamber **9** is 0° C. and a set temperature of the second accommodation chamber **10** is 60° C.

(134) In a case where a set temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** is within 60° C., if the set temperature of the first accommodation chamber **9** is switched from 0° C. to -10° C., the microcomputer **81** switches a temperature settable range of the second accommodation chamber **10** from -18° C. to 60° C. to -18° C. to 50° C. In this case, the changed temperature settable range may be displayed on the left room temperature display part **61f** that is a display part of the second accommodation chamber **10**. Consequently, a user can set a temperature after recognizing the temperature settable range. Even if the left room temperature setting button **63** is operated such that the set temperature of the second accommodation chamber **10** exceeds 50° C. (exceeds the temperature settable range), the left room temperature display part **61f** does not display the set temperature that exceeds the temperature settable range. In other words, even if an operation of increasing the set temperature from 50° C. is operated, the display of the set temperature remains at 50° C. A set temperature difference is not limited to 60° C. For example, it is possible to perform switching control on the basis of a

temperature of the outside air or a remaining battery level.

(135) Alternatively, control may be performed such that display of a set temperature deviating from the temperature settable range is permitted, but an operation cannot be started even if the execution button **67** is pressed. In this case, it is preferable to notify a user that the set temperature deviates from the temperature settable range by increasing a blinking speed of the set temperature display or displaying an error. The notification may be performed at a timing at which the execution button **67** is pressed, or at a timing in a case where display of the set temperature deviating from the temperature settable range is started. In a case where the set temperature of one of the first accommodation chamber **9** and the second accommodation chamber **10** is changed and thus the temperature settable range of the other becomes narrower than normal (18°C. to 60°C.), the narrowed temperature settable range may be displayed on the temperature display part on the other in addition to the set temperature (not illustrated).

(136) As illustrated in (A) of FIG. **33**, in a case where the power button **65** is pressed to start the electric device **1**, the operation mode is switched to the two-room mode that is an initial operation mode, and the right room temperature display part **61e** and the left room temperature display part **61f** display an initial set temperature of 10°C. in a blinking manner. After three seconds (after a predetermined time), the right room temperature display part **61e** and the left room temperature display part **61f** display the current temperature in a lighting manner. In a case where either the right room temperature setting button **62** or the left room temperature setting button **63** is pressed, the right room temperature display part **61e** and the left room temperature display part **61f** display an initial set temperature of 10°C. in a blinking manner. The set temperature is displayed for 3 seconds (predetermined time) after the right room temperature setting button **62** or left room temperature setting button **63** is last pressed. In a case where the right room temperature setting button **62** or left room temperature setting button **63** is pressed during that time, the set temperature is changed.

(137) In a case where the lower part of the right room temperature setting button **62** is pressed three times while the right room temperature display part **61e** and the left room temperature display part **61f** are displaying an initial set temperature of 10°C. in a blinking manner, the set temperature display on the right room temperature display part **61e** is changed to -5°C. After that, in a case where the upper part of the left room temperature setting button **63** is pressed ten times, the set temperature display of the left room temperature display part **61f** is changed to 60°C. , and the set temperature display of the right room temperature display part **61e** is changed to 0°C. due to control for making the set temperature difference within 60°C. After three seconds, the right room temperature display part **61e** and the left room temperature display part **61f** display the current temperatures in a lighting manner.

(138) In a case where the mode switching button **64** is pressed, the operation mode is switched to the large room single mode, the set temperature display of the right room temperature display part **61e** is changed to 10°C. , and the left room temperature display part **61f** is lighted off. After three seconds, the right room temperature display part **61e** and the left room temperature display part **61f** display the current temperatures in a lighting manner. In a case where the mode switching button **64** is pressed again, the operation mode is switched to the small room single mode, the right room temperature display part **61e** is lighted off, and the set temperature display of the left room temperature display part **61f** is changed to 10°C. After three seconds, the right room temperature display part **61e** and the left room temperature display part **61f** display the current temperatures in a lighting manner. In a case where the mode switching button **64** is pressed again, the operation mode is switched to the one-room mode, the set temperature display of the right room temperature display part **61e** is changed to 10°C. , the left room temperature display part **61f** is lighted off, and the partition line between the right room temperature display part **61e** and the left room temperature display part **61f** is lighted off.

(139) FIG. **34** is a circuit block diagram of the electric device **1**. In this drawing, two battery packs

29 are distinguished as battery packs 29a and 29b. ADC power supply 90 is, for example, an AC adapter, is connected to an external AC power supply (not illustrated), converts AC power into DC power (for example, DC 12 V), and supplies the DC power to the power input terminal 28 of the electric device 1. Alternatively, the DC power supply 90 is, for example, an in-vehicle power supply (in-vehicle battery), and supplies DC power to the power input terminal 28. The power input terminal 28 also functions as an in-vehicle power supply connection part to which an in-vehicle power supply can be connected. A voltage of the in-vehicle power supply is, for example, about DC 12.5 V to 14 V, and varies depending on a state of the in-vehicle power supply, a length of a power cord, and the like. Similarly, the magnitude of the voltage drop when a current is output from the in-vehicle power supply also varies depending on a state of the in-vehicle power supply, a length of a power cord, and the like. As the deterioration in the in-vehicle power supply progresses, the internal resistance increases and the voltage drop during current output also increases. A compressor drive circuit 48 is provided on the compressor 41 side.

(140) The electric device 1 includes, on the control circuit board 80, a microcomputer 81 that is an operation control part, a microcomputer 82 that is a charging control part, a control power supply 83, a rotation speed setting circuit 84, a shunt resistor 85, battery voltage detection circuits 86a and 86b, a DC power supply voltage detection circuit 86c that is a state detection part, a charging circuit 88, and a shunt resistor 89. The microcomputers 81 and 82 function as control parts that control the supply of power to the compressor 41. The microcomputers 81 and 82 may not be separate from each other, and may be a single microcomputer (microcontroller). The control power supply 83 converts an input voltage from the DC power supply 90 or the battery pack 29a or 29b into a power supply voltage (for example, 5V) for the microcomputers 81 and 82, and the like, and supplies the power supply voltage to the microcomputers 81 and 82, and the like. The battery voltage detection circuits 86a and 86b transmit detection signals respectively corresponding to the voltages of the battery packs 29a and 29b to the microcomputer 82. The DC power supply voltage detection circuit 86c transmits a detection signal corresponding to the voltage of the DC power supply 90 to the microcomputer 81.

(141) The microcomputer 81 controls the entire operation related to cooling and heating of the electric device 1. The microcomputer 81 controls on/off of a switching element Q3 provided in a current path of the compressor drive circuit 48, and controls driving and stopping of the compressor 41. The microcomputer 81 transmits a rotation speed determination signal to the compressor drive circuit 48 via the rotation speed setting circuit 84 to control a rotation speed of the compressor 41. As illustrated in FIG. 35, the rotation speed setting circuit 84 has resistors R1 to R3 having different resistance values connected in parallel between rotation speed determination terminals of the compressor drive circuit 48, and switching elements Q6 to Q8 respectively connected in series to the resistors R1 to R3. The switching elements Q6 to Q8 are alternatively turned on by a rotation speed selection signal from the microcomputer 81, for example. Depending on which of the switching elements Q6 to Q8 is turned on, a voltage between the rotation speed determining terminals of the compressor drive circuit 48 changes, and a set rotation speed of the compressor 41 is determined.

(142) The microcomputer 81 receives an operation on the setting part 60 as an electrical signal, and controls display on the setting part 60 (display on the display part 61). The microcomputer 81 controls opening and closing of the first regulating valve 47a and the second regulating valve 47b to control a flow of the refrigerant in the first refrigerant pipe and the second refrigerant pipe 46. The microcomputer 81 controls turning-on and turning-off of switching elements Q4 and Q5 respectively provided in current paths of the first heating part 51 and the second heating part 52 to control driving of the first heating part 51 and the second heating part 52. The microcomputer 81 detects temperatures (current temperatures) of the first accommodation chamber 9 and the second accommodation chamber 10 on the basis of the output signals from the first thermistor 55 and the second thermistor 56. The microcomputer 81 detects a drive current for the compressor 41 and

drive currents of the first heating part **51** and the second heating part **52** on the basis of a voltage of the shunt resistor **85**. The shunt resistor **85** is a block of resistors respectively connected in series to the switching elements **Q3** to **Q5**. The microcomputer **81** detects a voltage of the DC power supply **90** on the basis of a detection signal from the DC power supply voltage detection circuit **86c**.

(143) The microcomputer **82** controls charging of the battery packs **29a** and **29b** in the electric device **1**. The microcomputer **82** controls a charging voltage through control of the charging circuit **88**. Under the control of the microcomputer **82**, the charging circuit **88** converts the input voltage from the DC power supply **90** into a charging voltage of the battery pack **29a** or **29b**, and supplies the charging voltage to the battery pack **29a** or **29b** (charging the battery pack **29a** or **29b**). The microcomputer **82** controls turning-on and turning-off of switching elements **Q1** and **Q2** provided between the output terminal of the charging circuit **88** and the charging terminals (C+terminals) of the battery packs **29a** and **29b** to determine which of the battery packs **29a** and **29b** is to be charged. Diodes **D5** and **D6** for backflow prevention are connected between the switching elements **Q1** and **Q2** and the microcomputer **82**. The microcomputer **82** detects voltages of the battery packs **29a** and **29b** on the basis of detection signals from the battery voltage detection circuits **86a** and **86b**. The microcomputer **82** detects a charging current on the basis of a voltage of the shunt resistor **89** provided in the output current path of the charging circuit **88**.

(144) In addition to charging control, the microcomputer **82** controls turning-on and turning-off of relays **S1** and **S2** that are switches connected to the positive terminals (+terminals) of the battery packs **29a** and **29b** to determine which of the battery packs **29a** and **29b** is to be discharged. In a case where power is supplied from the DC power supply **90**, the microcomputer **82** turns off the relays **S1** and **S2** such that the battery packs **29a** and **29b** are not discharged. Fuses **F1** to **F3** and diodes **D1** to **D3** for backflow prevention are respectively connected to the positive terminals of the battery packs **29a** and **29b** and the DC power supply **90**. The microcomputers **81** and **82** can communicate with each other and share various types of information. For example, the microcomputer **82** may acquire voltage information regarding the DC power supply **90** by communicating with the microcomputer **81**.

(145) (A) of FIG. **36** is a time chart illustrating power supply selection, a voltage of the DC power supply **90**, and a drive current for the compressor in a case where the DC power supply **90** is a normal in-vehicle power supply. The normal in-vehicle power supply is an in-vehicle power supply in which deterioration has not progressed, and is an in-vehicle power supply in a state of having a sufficient capacity necessary for starting the compressor **41**. Immediately after the start of operation at time point **t11**, a large starting current temporarily flows and the voltage of DC power supply **90** drops. However, the voltage of the DC power supply **90** is not lower than 11 V that is a threshold value for switching a power supply source to the compressor **41** (hereinafter, a “power supply source”) from the DC power supply **90** to the battery pack **29a** or **29b** (hereinafter, a “threshold value for switching to the battery”), and it is possible to start the compressor **41**, that is, to operate the electric device **1**. After a large starting current temporarily flows, the drive current for the compressor **41** decreases to a value corresponding to a load. Along with this, the voltage of the DC power supply **90** also increases.

(146) (B) of FIG. **36** is a time chart illustrating power supply selection, a voltage of the DC power supply **90**, and a drive current for the compressor **41** in a case where the DC power supply **90** is an in-vehicle power supply in which deterioration has progressed in an operation of the electric device **1** in a comparative example. The in-vehicle power supply in which deterioration has progressed has a lower voltage than that of a normal in-vehicle power supply, and the voltage drop is large in a case where a current flows. The microcomputer **82** first selects the DC power supply **90** as a power supply source (turns off the relays **S1** and **S2**). Immediately after the start of operation at time point **t11**, a large starting current temporarily flows and the voltage of DC power supply **90** drops. In this case, the voltage of the DC power supply **90** becomes lower than 11 V that is a threshold value for switching to the battery. As a result, the microcomputer **81** determines that the electric device **1**

cannot be operated with the power of the DC power supply **90** and starts a process of stopping the compressor **41**. The microcomputer **82** switches a power supply source to the battery pack **29a** or **29b**. The relays **S1** and **S2** correspond to a switching circuit in the present invention.

(147) At time point **t13**, the process of stopping the compressor **41** is completed, and the drive current for the compressor **41** becomes 0. In this case, the voltage of the DC power supply **90** recovers to a value higher than 12 V that is a threshold value for switching a power supply source from the battery pack **29a** or **29b** to the DC power supply **90** (hereinafter, referred to as a “threshold value for switching to the in-vehicle power supply”). Therefore, the microcomputer **82** switches a power supply source from the battery pack **29a** or **29b** to the DC power supply **90**, and attempts to start the compressor **41** again by using the power of the DC power supply **90** at time point **t15**. However, since the state of the DC power supply **90** has not changed, the compressor **41** cannot be started (the electric device **1** cannot be operated), in the same manner as in the case of the start of operation at time point **t11**. An interval between time points **t13** and **t15** is a waiting time required for protecting the compressor **41**, for example, two minutes. Since the compressor **41** discharges the refrigerant, if the compressor **41** is restarted immediately after being stopped, the mechanism of the compressor **41** may be adversely affected depending on a state of the refrigerant. Therefore, a waiting time is provided from when the compressor **41** is stopped to when the compressor **41** is restarted.

(148) As described above, in a case where a power supply source is switched to the DC power supply **90** only under the condition that the voltage of the DC power supply **90** is higher than 12 V, a circulation of a failure to start the compressor **41** with the power of the DC power supply **90**, switching a power supply source to the battery pack **29a** or **29b**, switching a power supply source to the DC power supply **90** due to the voltage recovery of the DC power supply **90**, and a failure to start the compressor **41** with the power of the DC power supply **90** may occur such that driving of the compressor **41**, that is, an operation of the electric device **1** cannot be substantially performed, and thus a cooling failure may be caused. FIG. **37** illustrates an operation of the electric device **1** for solving such problems.

(149) FIG. **37** is a time chart illustrating power supply selection, a voltage of the DC power supply **90**, a voltage of the battery pack **29a** or **29b**, a voltage input to the compressor drive circuit **48**, and a drive current for the compressor **41** in a case where the DC power supply **90** is an in-vehicle power supply in which deterioration has progressed in an operation of the electric device **1** of the embodiment.

(150) In a period A immediately after the power button **65** is pressed, the microcomputer **82** selects the DC power supply **90** as a power supply source (the relays **S1** and **S2** are turned off). In a period B, the microcomputer **81** starts the compressor **41** with the power of the DC power supply **90**, but a starting current for the compressor **41** causes the voltage of the DC power supply **90** to drop below an operation stop threshold value (for example, 11 V), and thus the compressor **41** is stopped. The microcomputer **82** determines that the compressor **41** cannot be started by using the DC power supply **90** (determines that the DC power supply **90** cannot be used), and switches a power supply source to the battery pack **29a** or **29b** at time point C (turns on the relay **S1** or **S2**). The fact that the voltage of the DC power supply **90** decreases below the operation stop threshold value due to the starting current for the compressor **41** corresponds to an operation stop condition in the present invention.

(151) During a period D after a waiting time necessary for protecting the compressor **41** has passed from stoppage of the compressor **41** during the period B, the microcomputer **81** starts the compressor **41** with the power of the battery pack **29a** or **29b**. Since the voltage of the battery pack **29a** or **29b** is high, the compressor **41** is successfully started, and the compressor **41** is driven with the power of the battery pack **29a** or **29b** for a predetermined time regardless of the voltage of the DC power supply **90**. During this time, a voltage of the in-vehicle power supply increases due to, for example, traveling of a vehicle equipped with the in-vehicle power supply. At time point E after

the predetermined time has elapsed, the microcomputer **81** temporarily stops driving the compressor **41**. Since there is a possibility that a state of the in-vehicle power supply has changed and the electric device is operable, the microcomputer **82** switches a power supply source to the DC power supply **90** again at time point F. Elapse of a predetermined time corresponds to an operation resuming condition in the present invention. If the voltage of the DC power supply **90** has not increased to the operation resuming threshold value (for example, 12 V) or higher after a predetermined time has elapsed, the compressor **41** may be continued to be driven with the power of the battery pack **29a** or **29b** (the compressor **41** is not stopped). In this case, the operation resuming condition is that a predetermined time has elapsed and the voltage of the DC power supply **90** exceeds the operation resuming threshold value.

(152) At time point G after the waiting time necessary for protecting the compressor **41** has elapsed from time point E, the microcomputer **81** starts the compressor **41** with the power of the DC power supply **90**. The voltage of the DC power supply **90** has increased compared to the previous start using the power of the DC power supply **90**, and the voltage of the DC power supply **90** does not fall below the operation stop threshold value during the present starting, and the compressor **41** can be successfully started. In a case where the voltage of the DC power supply **90** decreases below the operation stop threshold value during the present starting, the compressor **41** is stopped in the same manner as in the previous starting, the compressor **41** is driven for a predetermined time with the power of the battery pack **29a** or **29b** and then stopped. After that, the compressor **41** is tried to be started again with the power of the DC power supply **90**.

(153) As described above, in a state in which an in-vehicle power supply as the DC power supply **90** is connected to the power input terminal **28** and the battery pack **29a** or **29b** is connected to the battery pack attachment part **22a**, in a case where a power supply source is switched from the in-vehicle power supply to the battery pack **29a** or **29b** according to a detection result from the DC power supply voltage detection circuit **86c**, the microcomputer **82** maintains the supply of power to the compressor **41** from the battery pack **29a** or **29b** until a predetermined time elapses regardless of whether the voltage of the DC power supply **90** is higher than the threshold value for switching to the in-vehicle power supply. As a result, it is possible to prevent that the compressor **41** cannot be driven, that is, the electric device **1** cannot be substantially operated, and thus a cooling failure can be suppressed. That is, the problem in the operation in the comparative example illustrated in (B) of FIG. **36** can be solved favorably. The effect of being able to prevent that the electric device cannot be substantially operated is the same even in a case where an output part is not the compressor **41** (for example, a motor other than the compressor) or in a case where the electric device is not a cold-hot storage. The in-vehicle power supply connection part **28** corresponds to a first power supply source in the present invention, and the battery pack attachment part **22a** corresponds to a second power supply source in the present invention.

(154) The predetermined time is longer than the waiting time required for protecting the compressor **41**. Therefore, an interval for determining whether or not the DC power supply **90** can be used on the basis of a detection result from the DC power supply voltage detection circuit **86** is longer than the waiting time required for protecting the compressor **41**.

(155) The predetermined time may be a time until a user performs an operation of stopping the compressor **41**, that is, a time until the user presses the power button **65**. In other words, in a case where the microcomputer **82** determines that the DC power supply **90** cannot be used on the basis of the detection result from the DC power supply voltage detection circuit **86**, the supply of power from the DC power supply **90** to the compressor **41** may be prohibited until the user performs an operation of stopping the compressor **41**. In this case, in a case where it is detected that a plug is inserted into or detached from the power input terminal **28**, an attempt may be made to start the compressor **41** with the power of the DC power supply **90**.

(156) In a case where the DC power supply **90** is connected to the power input terminal **28** but a power supply source is not the DC power supply **90**, the microcomputer **81** may notify a user by

causing the external power supply connection display part **61b** or the like to blink. As a result, the user can recognize that the electric device **1** cannot be operated because the in-vehicle power supply used as the DC power supply **90** is not in good condition, or that the battery pack **29a** or **29b** is a power supply source, and thus convenience is high.

(157) As a modification example of the operation in FIG. **37**, in a case where the microcomputer **82** determines that the DC power supply **90** cannot be used on the basis of a detection result from the DC power supply voltage detection circuit **86**, a power supply source may be switched to the battery pack **29a** or **29b** from the DC power supply **90** without stopping the driving of the compressor **41**. Specifically, at the end of the period B, the relay **51** or **S2** may be turned on without stopping the driving of the compressor **41**, a power supply source may be switched to the battery pack **29a** or **29b**, and the driving of the compressor **41** may be continued.

(158) FIG. **38** is a flowchart illustrating a main routine in the electric device **1**. In a case where the power button **65** is pressed, the control power supply **83** is activated and the microcomputers **81** and **82** are started (**S1**). The microcomputers **81** and **82** execute a power selection routine illustrated in FIG. **39** (**S2**). The microcomputer **81** reads a set temperature selected by the setting part **60** (**S5**), reads output signals (the first accommodation chamber **9** and the second accommodation chamber **10**) from the first thermistor **55** and the second thermistor **56** (**S6**), and executes control in an operation mode selected by the setting part **60** (any of **S7** to **S10**).

(159) FIG. **39** is a flowchart illustrating the power selection routine (**S2**) in FIG. **38**. In a case where power is supplied from the DC power supply **90** (Yes in **S2a**) and at least one of the battery packs **29a** and **29b** is connected (Yes in **S2b**), the microcomputer **82** executes a DC power state determination routine (**S3**). In a case where a battery priority flag is not in an ON state (No in **S2c**), the microcomputer **82** executes a charging routine illustrated in FIG. **41** (**S4**) and sets the power supply source to the DC power supply **90** (**S2d**). In a case where the battery priority flag is in an ON state (Yes in **S2c**), the microcomputer **82** sets the power supply source to the battery pack **29a** or **29b** (**S2e**). In a case where power is not supplied from the DC power supply **90** (No in **S2a**), the microcomputer **82** sets the power supply source to the battery pack **29a** or **29b** (**S2e**). In a case where neither of the battery packs **29a** and **29b** are connected (No in **S2b**), the microcomputer **82** sets the power supply source to the DC power supply **90** (**S2d**). In a case where the power supply source is determined, the power selection routine is ended.

(160) FIG. **40** is a flowchart illustrating a DC power state determination routine (**S3**) in FIG. **39**. The power state determination routine is a routine for setting an ON state and an OFF state of the battery priority flag. The battery priority flag is in an OFF state in an initial state. That is, the battery priority flag is set to an OFF state in a case where the power supply is turned of.

(161) In a case where the battery priority flag is not in an ON state (No in **S3a**), when the voltage of the DC power supply **90** decreases below the threshold value for switching to the battery during starting of the motor of the compressor **41** (Yes in **S3b**), the microcomputer **81** sets the battery priority flag to an ON state (**S3c**), and ends the DC power state determination routine. In a case where the voltage of the DC power supply **90** does not decrease below the threshold value for switching to the battery during starting of the motor of the compressor **41** in **S3b** (No in **S3b**), the microcomputer **81** ends the DC power state determination routine without changing the battery priority flag from an OFF state.

(162) In a case where the battery priority flag is in an ON state (Yes in **S3a**), when 15 minutes that is an example of a predetermined time has elapsed from setting of the battery priority flag to an ON state (Yes in **S3d**), the microcomputer **81** sets the battery priority flag to an OFF state (**S3e**), and ends the DC power state determination routine. In a case where 15 minutes has not elapsed from setting of the battery priority flag to an ON state in **S3d** (No in **S3d**), the microcomputer **81** ends the DC power state determination routine without changing the battery priority flag from an ON state.

(163) The condition for proceeding to Yes in **S3b** and setting the battery priority flag to an ON state

may be that a drive current for the compressor **41** cannot be detected instead of or in addition to a voltage of the DC power supply **90** decreasing below the threshold value for switching to the battery. In this case, it is detected that the motor of the compressor **41** cannot be started on the basis of a drive current for the compressor **41** and the battery priority flag is set to an ON state. In this case, the battery priority flag may be set to an ON state in a case where a state in which the drive current for the compressor **41** cannot be detected is detected for a predetermined time (for example, 30 seconds). Alternatively, the condition for proceeding to Yes in **S3b** and setting the battery priority flag to an ON state may be that the drive current for the compressor **41** is too large (exceeds a predetermined value). If the voltage of the DC power supply **90** is low, a current becomes large to secure power. Therefore, the fact that the drive current for the compressor **41** is excessive indicates that the voltage of the DC power supply **90** has not reached a required value. (164) The condition for proceeding to Yes in **S3d** and setting the battery priority flag to an OFF state may be that the voltage of the DC power supply **90** has increased to a voltage that can withstand starting of the compressor **41** (a predetermined voltage higher than the threshold value for switching to the in-vehicle power supply) or higher. This voltage is high enough not to fall below the threshold value for switching to the battery, for example, 15.5 V, even if a voltage drop occurs due to a starting current for the compressor **41**.

(165) FIG. **41** is a flowchart illustrating a specific operation of the charging routine (**S4** in FIG. **38**) in the electric device **1**. The microcomputer **82** selects a connection port (port **1**) of the battery pack **29a** (**S11**), and if the battery pack **29a** can be charged (Yes in **S12**), charges the battery pack **29a** (**S13**). In a case where the charging of the battery pack **29a** is completed (Yes in **S14**), the microcomputer **82** selects a connection port (port **2**) of the battery pack **29b** (**S15**), and if the battery pack **29b** can be charged (Yes in **S16**), charges the battery pack **29b** (**S17**). The microcomputer **82** recognizes that the charged battery packs **29a** and **29b** can be discharged (**S19**).

(166) FIG. **42** is a flowchart illustrating a specific operation of the electric device **1** in the large room single mode (**S7** in FIG. **38**). The microcomputer **81** reads a set temperature of the first accommodation chamber **9** (**S21**) and reads an output signal from the first thermistor **55** (the current temperature of the first accommodation chamber **9**) (**S22**). In a case where the current temperature of the first accommodation chamber **9** is higher than the set temperature+2° C. (Yes in **S23**), the microcomputer **81** sets a rotation speed of the compressor **41** to 2,500 rpm (**S24**). The microcomputer **81** turns on (opens) the first regulating valve **47a** and turns off (closes) the second regulating valve **47b** (**S25**), and then drives the compressor **41** (**S26**). In a case where the current temperature of the first accommodation chamber **9** is not higher than the set temperature+2° C. in **S23** (No in **S23**), if the current temperature of the second accommodation chamber **10** is lower than the set temperature—2° C. (Yes in **S27**), the microcomputer **81** drives the first heating part **51** and stops the second heating part **52** (**S28**).

(167) FIG. **43** is a flowchart illustrating a specific operation in the small room single mode (**S8** in FIG. **38**) of the electric device **1**. The microcomputer **81** reads a set temperature of the second accommodation chamber **10** (**S31**), and reads an output signal from the second thermistor **56** (the current temperature of the second accommodation chamber **10**) (**S32**). In a case where the current temperature of the second accommodation chamber **10** is higher than the set temperature+2° C. (Yes in **S33**), the microcomputer **81** sets a rotation speed of the compressor **41** to 2,000 rpm (**S34**). The microcomputer **81** turns on (opens) the second regulating valve **47b** and turns off (closes) the first regulating valve **47a** (**S35**), and then drives the compressor **41** (**S36**). In a case where the current temperature of the second accommodation chamber **10** is not higher than the set temperature+2° C. in **S33** (No in **S33**), if the current temperature of the first accommodation chamber **9** is lower than the set temperature—2° C. (Yes in **S37**), the microcomputer **81** stops the first heating part **51** and drives the second heating part **52** (**S38**).

(168) FIG. **44** is a flowchart illustrating a specific operation of the electric device **1** in the one-room mode (**S9** in FIG. **38**). The microcomputer **81** reads a set temperature in the one-room mode (**S41**),

and reads an output signal from the first thermistor **55** (the current temperature of the first accommodation chamber **9**) (**S42**). In a case where the current temperature of the first accommodation chamber **9** is higher than the set temperature+2° C. (Yes in **S43**), the microcomputer **81** sets a rotation speed of the compressor **41** to 3,000 rpm (**S44**). The microcomputer **81** turns on (opens) the first regulating valve **47a** and the second regulating valve **47b** (**S45**), and then drives the compressor **41** (**S46**). In a case where the current temperature of the first accommodation chamber **9** is not higher than the set temperature+2° C. in **S43** (No in **S43**), if the current temperature of the first accommodation chamber **9** is lower than the set temperature—2° C. (Yes in **S47**), the microcomputer **81** drives the first heating part **51** and the second heating part **52** (**S48**). In **S42**, **S43**, and **S47**, an output signal from the second thermistor **56** (the current temperature of the second accommodation chamber **10**) may be used as the current temperature, or an average of the current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** may be calculated from both of output signals from the first thermistor **55** and the second thermistor **56** and used.

(169) FIG. **45** is a flowchart illustrating a specific operation of the electric device **1** in the two-room mode (**S10** in FIG. **38**). The microcomputer **81** reads set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** (**S51**), and reads output signals from the first thermistor **55** and the second thermistor **56** (the current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**) (**S52**). In a case where the current temperature of the first accommodation chamber **9** is higher than the set temperature+2° C. (Yes in **S53**) and the current temperature of the second accommodation chamber **10** is higher than the set temperature+2° C. (Yes in **S54**), the microcomputer **81** sets a rotation speed of the compressor **41** to 3,000 rpm (**S55**). The microcomputer **81** turns on (opens) the first regulating valve **47a** and the second regulating valve **47b** (**S56**), and then drives the compressor **41** (**S57**). In a case where the current temperature of the second accommodation chamber **10** is not higher than the set temperature+2° C. in **S54** (No in **S54**), the microcomputer **81** sets the rotation speed of the compressor **41** to 2,500 rpm (**S58**). The microcomputer **81** turns on (opens) the first regulating valve **47a** and turns off (closes) the second regulating valve **47b** (**S59**), and then drives the compressor **41** (**S60**). In a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature—2° C. (Yes in **S74**), the microcomputer **81** stops the first heating part **51** and drives the second heating part **52** (**S75**).

(170) In a case where the current temperature of the first accommodation chamber **9** is not higher than the set temperature+2° C. in **S53** (No in **S53**), and in a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature—2° C. (Yes in **S61**) and the current temperature of the second accommodation chamber **10** is higher than the set temperature+2° C. (**S62**), the microcomputer **81** sets the rotation speed of the compressor **41** to 2,000 rpm (**S63**). The microcomputer **81** turns off (closes) the first regulating valve **47a** and turns on (opens) the second regulating valve **47b** (**S64**), and then drives the compressor **41** (**S65**). On the other hand, the microcomputer **81** drives the first heating part **51** and stops the second heating part **52** (**S66**). In a case where the current temperature of the second accommodation chamber **10** is not higher than the set temperature+2° C. in **S62** (No in **S62**), if the current temperature of the second accommodation chamber **10** is lower than the set temperature—2° C. (Yes in **S67**), the microcomputer **81** drives the first heating part **51** and the second heating part **52** (**S68**). In a case where the current temperature of the second accommodation chamber **10** is not lower than the set temperature—2° C. in **S67**, the microcomputer **81** drives the first heating part **51** and stops the second heating part **52** (**S69**).

(171) In a case where the current temperature of the first accommodation chamber **9** is not lower than the set temperature—2° C. in **S61** (No in **S61**), if the current temperature of the second accommodation chamber **10** is higher than the set temperature+2° C. (Yes in **S70**), and the microcomputer **81** sets the rotation speed of the compressor **41** to 2,000 rpm (**S71**). The

microcomputer **81** turns off (closes) the first regulating valve **47a** and turns on (opens) the second regulating valve **47b** (**S72**), and then drives the compressor **41** (**S73**). In a case where the current temperature of the second accommodation chamber **10** is not higher than the set temperature+2° C. in **S70** (No in **S70**), if the current temperature in the second accommodation chamber **10** is lower than the set temperature—2° C. (Yes in **S74**), the microcomputer **81** stops the first heating part **51** and drives the second heating part **52** (**S75**).

(172) In the cooling control for the first accommodation chamber **9**, the microcomputer **81** stops turning-on the first regulating valve **47a** and driving the compressor **41** in a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature by a predetermined value (for example, 2° C.). In order to prevent the first heating part **51** from being driven immediately after stopping turning-on of the first regulating valve **47a** and driving of the compressor **41**, a waiting time may be provided between stopping of turning-on of the first regulating valve **47a** and driving of the compressor **41** and determination of whether or not to drive the first heating part **51**.

(173) Similarly, in the cooling control for the second accommodation chamber **10**, the microcomputer **81** stops turning-on of the second regulating valve **47b** and driving of the compressor **41** in a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature by a predetermined value (for example, 2° C.). In order to prevent the second heating part **52** from being driven immediately after stopping turning-on of the second regulating valve **47b** and driving of the compressor **41**, a waiting time may be provided between stopping of turning-on of the second regulating valve **47b** and driving of the compressor **41** and determination of whether or not to drive the second heating part **52**.

(174) In the heating control for the first accommodation chamber **9**, the microcomputer **81** stops driving the first heating part **51** in a case where the current temperature of the first accommodation chamber **9** is higher than the set temperature by a predetermined value (for example, 2° C.). In order to prevent the cooling control for the first accommodation chamber **9** from being started immediately after the first heating part **51** is stopped, a waiting time may be provided between stopping of the first heating part **51** and determination of whether or not to start the cooling control for the first accommodation chamber **9**.

(175) Similarly, in the heating control for the second accommodation chamber **10**, the microcomputer **81** stops driving the second heating part **52** in a case where the current temperature of the second accommodation chamber **10** is higher than the set temperature by a predetermined value (for example, 2° C.). In order to prevent the cooling control for the second accommodation chamber **10** from being started immediately after the second heating part **52** is stopped, a waiting time may be provided between stopping of the second heating part **52** and determination of whether or not to start the cooling control for the second accommodation chamber **10**.

(176) In FIGS. **42** to **45**, 2,500 rpm is an example of the maximum drive intensity (first intensity) of the compressor **41** according to a size of the first accommodation chamber **9**. 2,000 rpm is an example of the maximum drive intensity (second intensity) of the compressor **41** according to a size of the second accommodation chamber **10**. 3,000 rpm is an example of the maximum drive intensity (third intensity) of the compressor **41** corresponding to a total size of the first accommodation chamber **9** and the second accommodation chamber **10**. That is, the microcomputer **81** switches drive states (maximum drive intensity) of the compressor **41** according to a size of the accommodation chamber to be cooled. In a case of increasing a drive state, the microcomputer **81** may perform control of switching between an opening state and a closing state of the regulating valve **87** and switching between drives states, and in a case of decreasing the drive state, the microcomputer **81** may perform control of switching between drive states and then switching between an opening state and a closing state of the regulating valve **87**.

(177) FIG. **46** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in

a case where an operation is performed in the two-room mode when a set temperature of the first accommodation chamber **9** is 10° C. and a set temperature of the second accommodation chamber **10** is -18° C. At the beginning of operation, temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** are about 30° C., the control in S55 to S57 in FIG. 45 is performed, and the compressor **41** is driven at 3,000 rpm. As a result, the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** decrease. In a case where the temperature of the first accommodation chamber **9** reaches 8° C. around 8 minutes, the control in S71 to S73 in FIG. 45 is performed, the first accommodation chamber **9** is stopped to be cooled (transition to cooling of only the second accommodation chamber **10**), and the rotation speed of the compressor **41** decreases to 2,000 rpm. In a case where the temperature of the first accommodation chamber **9** exceeds 12° C. around 15 minutes, the control in S55 to S57 in FIG. 45 is performed, the cooling of the first accommodation chamber **9** is resumed, and the rotation speed of the compressor **41** increases to 3,000 rpm. Thereafter, the same operation is repeatedly performed, the temperature of the first accommodation chamber **9** is kept around 10° C., and the temperature of the second accommodation chamber **10** decreases toward -18° C.

(178) FIG. 47 is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where an operation is performed in the two-room mode when a set temperature of the first accommodation chamber **9** is 18° C. and a set temperature of the second accommodation chamber **10** is 10° C. At the beginning of operation, temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** are about 30° C., the control in S55 to S57 in FIG. 45 is performed, and the compressor **41** is driven at 3,000 rpm. As a result, the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** decrease. In a case where the temperature of the second accommodation chamber **10** reaches 8° C. around 8 minutes, the control in S58 to S60 in FIG. 45 is performed, the second accommodation chamber **10** is stopped to be cooled (transition to cooling of only the first accommodation chamber **9**), and the rotation speed of the compressor **41** decreases to 2,500 rpm. In a case where the temperature of the second accommodation chamber **10** exceeds 12° C. around 15 minutes, the control in S55 to S57 in FIG. 45 is performed, the cooling of the second accommodation chamber **10** is resumed, and the rotation speed of the compressor **41** increases to 3,000 rpm. Thereafter, the same operation is repeatedly performed, the temperature of the second accommodation chamber **10** is kept around 10° C., and the temperature of the first accommodation chamber **9** decreases toward -18° C.

(179) FIG. 48 is a time chart illustrating a voltage input to the compressor **41**, a rotation speed of the compressor **41**, opening and closing signals for the first regulating valve **47a** and the second regulating valve **47b**, turning-on and turning-off of the first heating part **51** and the second heating part **52**, and temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**. The first regulating valve **47a** and the second regulating valve **47b** are solenoid valves that apply a pulse signal only when switching between opening and closing.

(180) At time point t_0 , temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** are 30° C., and the microcomputer **81** performs the control in S55 to S57 in FIG. 45. That is, the microcomputer **81** outputs a valve opening/closing signal for setting both the first regulating valve **47a** and the second regulating valve **47b** to an open state, and then drives the compressor **41** at 3,000 rpm. In a case where the temperature of the second accommodation chamber **10** reaches 8° C. at time point t_1 , the microcomputer **81** performs the control in S58 to S60 in FIG. 45. That is, the microcomputer **81** outputs a valve opening/closing signal for opening the first regulating valve **47a** and closing the second regulating valve **47b**, and then drives the compressor **41** at 2,500 rpm. In a case where the temperature of the second accommodation chamber **10** reaches 12° C. at time point t_2 , the microcomputer **81** performs the control in S55 to S57 in FIG. 45. That is, the microcomputer **81** outputs a valve opening/closing signal for setting both the first regulating valve **47a** and the second regulating valve **47b** to an open

state, and then drives the compressor **41** at 3,000 rpm.

(181) In a case where the temperature of the second accommodation chamber **10** reaches 8° C. at time point **t3**, the microcomputer **81** outputs a valve opening/closing signal for setting the first regulating valve **47a** to an open state and setting the second regulating valve **47b** to a closed state in the same manner as in the control at time point **t1**, and then drives the compressor **41** at 2,500 rpm. At time point **t3**, the cooling control for the second accommodation chamber **10** is temporarily stopped, but due to heat transfer to the first accommodation chamber **9** having a lower temperature, the temperature of the second accommodation chamber **10** also decreases after time point **t3**. At time point **t4** after a predetermined period of time has elapsed from time point **t3**, the microcomputer **81** starts driving the second heating part **52** (S75 in FIG. **45**). As a result, the temperature of the second accommodation chamber **10** begins to increase. In a case where the temperature of the second accommodation chamber **10** reaches 12° C. at time point **t5**, the microcomputer **81** stops the second heating part **52** and performs the control in S55 to S57 in FIG. **45**. That is, the microcomputer **81** outputs a valve opening/closing signal for setting both the first regulating valve **47a** and the second regulating valve **47b** to an open state, and then drives the compressor **41** at 3,000 rpm. The microcomputer **81** performs the same control as during time points **t3** to **t5** during time points **t6** to **t8**. According to the present embodiment, the following effects can be achieved.

(182) (1) Since the control circuit board **80** on which the charging circuit **88** is mounted and the fan **49** are accommodated in the second body part **4**, the charging circuit **88** can be efficiently cooled.

(183) (2) The fan air generated by the fan **49** is branched into the first air passage **33** for cooling the cooling mechanism **40** (especially the condenser **42**) and the second air passage **34** for cooling the control circuit board **80** by the branch part **32**. Therefore, the cooling mechanism **40** and the control circuit board **80** can be cooled separately and efficiently with fresh air.

(184) (3) By integrating the branch part **32** with the battery box **30**, the influence of the provision of the branch part **32** on the assembly process can be suppressed.

(185) (4) Since the first refrigerant pipe **45** is provided on the side surface of the first accommodation chamber **9** and the second refrigerant pipe **46** is provided on the side surface of the second accommodation chamber **10**, compared with a configuration in which the refrigerant pipes are disposed on the bottom surface, temperatures can be individually regulated over a wide area including the upper parts of the first accommodation chamber **9** and the second accommodation chamber **10**, and thus convenience can be improved. The first refrigerant pipe **45** is provided on three side surfaces of the first accommodation chamber **9**, and the second refrigerant pipe **46** is provided on three side surfaces of the second accommodation chamber **10**, and thus such an effect is remarkable.

(186) (5) Since the first refrigerant pipe **45** is provided with the first regulating valve **47a** and the second refrigerant pipe **46** is provided with the second regulating valve **47b**, flows of a refrigerant in the first refrigerant pipe **45** and the second refrigerant pipe **46** can be individually regulated, and temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** can be individually regulated. As a result, one of the first accommodation chamber **9** and the second accommodation chamber **10** can be set to be frozen, and the other can be set to be refrigerated, so that convenience can be improved.

(187) (6) Since the first heating part **51** for heating the first accommodation chamber **9** and the second heating part **52** for heating the second accommodation chamber **10** are provided, heating as well as cooling can be performed, and thus convenience is improved.

(188) One of the first accommodation chamber **9** and the second accommodation chamber **10** can be set to be frozen or refrigerated and the other can be set to be heated, or one can be set to be strongly heated and the other be set to be weakly heated, and thus convenience can be improved.

(189) (7) The microcomputer **81** operates the first heating part **51** or the second heating part **52** as necessary during the process of controlling the temperatures of the first accommodation chamber **9**

and the second accommodation chamber **10** to be lower than the outside air temperature. Thus, for example, in a case where a set temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** is large, and heat transfer to one of the first accommodation chamber **9** and the second accommodation chamber **10** having a lower set temperature causes the temperature of the other to further decrease below the set temperature, the temperature of the other can be prevented from deviating downward from the set temperature by heating the other.

(190) (8) The microcomputer **81** operates the cooling mechanism **40** as necessary even in the process of controlling the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** to be higher than the outside air temperature. Thus, for example, in a case where a set temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** is large, and heat transfer from one of the first accommodation chamber **9** and the second accommodation chamber **10** having the higher set temperature causes the temperature of the other to exceed the set temperature and further increase, by cooling the other, it is possible to prevent the temperature of the other from deviating upward from the set temperature.

(191) (9) Since the microcomputer **81** changes the maximum drive intensity of the compressor **41** according to a size of the accommodation chamber to be cooled, the compressor **41** can be driven with an intensity suitable for the size of the accommodation chamber, a load on the compressor **41** can be reduced, and the life of the compressor **41** can be extended. Specifically, for example, in a case where only the second accommodation chamber **10** is cooled, if a rotation speed of the compressor **41** is 2,500 rpm or 3,000 rpm, the rotation speed is too high (drive intensity is too high), and a phenomenon in which the refrigerant returns to the compressor **41** in a liquid state may occur. If the refrigerant returns to the compressor **41** in a liquid state, an extra load may be applied to the compressor **41** and thus the life of the compressor **41** may be shortened. According to the present embodiment, such problems can be suitably solved.

(192) (10) Since the first regulating valve **47a** and the second regulating valve **47b** are respectively provided in the portions of the first refrigerant pipe **45** and the second refrigerant pipe **46** where the refrigerant flows in a gaseous state, a refrigerant is sent out to the first refrigerant pipe **45** and the second refrigerant pipe **46** by the single compressor **41**, and thus it is possible to suppress unevenness in the flow of the refrigerant due to inclination even if the electric device **1** is in an inclined state. As a result, it is possible to prevent the refrigerant from unevenly flowing through either the first refrigerant pipe or the second refrigerant pipe **46**, and thus to suppress cooling of either the first accommodation chamber **9** or the second accommodation chamber **10** from being insufficient. Therefore, it is possible to prevent temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** from deviating from set temperatures.

(193) (11) The microcomputer **81** controls a set temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** to be within a predetermined value. Thus, it is possible to prevent the set temperature difference from exceeding the maximum temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** that can be realized through the heat insulation effect of the partition plate **70** (or the partition plate **170**). Consequently, it is possible to suppress the risk that the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** cannot reach the set temperatures.

(194) (12) In a case where a set temperature of one of the first accommodation chamber **9** and the second accommodation chamber **10** is changed such that a set temperature difference exceeds a predetermined value, the microcomputer **81** automatically changes a set temperature of the other such that the set temperature difference is included in the predetermined value, and thus convenience is high.

(195) (13) Since the partition plate **70** has a two-part configuration, it is more convenient to carry and store the partition plate than a single plate. Since the partition plate **170** is foldable, it is more

convenient to carry and store the partition plate than a single plate.

(196) (14) The partition plate **70** can be divided into the upper partition plate **71** and the lower partition plate **72**, and as illustrated in FIG. **26**, the lower partition plate **72** is attached and can also be used as a scattering prevention wall for the accommodation object **79** accommodated in the accommodation chamber **8**, and thus convenience is high.

(197) (15) The mating part **73** of the upper partition plate **71** and the lower partition plate **72** has a spigot structure, and the heat insulating material **74** extends to the inside of the spigot structure. Therefore, heat transfer between the first accommodation chamber **9** and the second accommodation chamber **10** can be suppressed. Similarly, the mating part **173** of the upper partition plate **171** and the lower partition plate **172** has a spigot structure, and the heat insulating material **174** extends to the inside of the spigot structure. Therefore, heat transfer between the first accommodation chamber **9** and the second accommodation chamber **10** can be suppressed.

(198) (16) The rail member **18** is configured to guide attachment of the partition plate **70** (or the partition plate **170**) via the recessed part **18b**, and does not protrude inward from the inner surface of the accommodation chamber **8**. Thus, it is possible to prevent a volume of the accommodation chamber **8** from decreasing due to the configuration for guiding the partition plate **70** (or the partition plate **170**). Although the rail member **18** protrudes outward from the outer surface of the accommodation chamber **8**, the first refrigerant pipe **45** and the second refrigerant pipe **46** are divided into left and right sides of the rail member **18** and wound around the accommodation chamber **8**. Therefore, the influence of the rail member **18** on the disposition of the first refrigerant pipe **45** and the second refrigerant pipe **46** is restricted.

(199) (17) In a state in which an in-vehicle power supply as the DC power supply **90** is connected to the power input terminal **28** and the battery pack **29a** or **29b** is connected to the battery pack attachment part **22a**, in a case where a power supply source is switched from the in-vehicle power supply to the battery pack **29a** or **29b** according to a detection result from the DC power supply voltage detection circuit **86c**, the microcomputer **82** maintains the supply of power from the battery pack **29a** or **29b** to the compressor **41** for a predetermined time. As a result, unlike a case where the power supply source is switched to the DC power supply **90** only on the condition that a voltage of the DC power supply **90** is higher than 12 V, it is possible to prevent that the compressor **41** cannot be driven, that is, the electric device **1** cannot be substantially operated due to repetition of unnecessary switching of a power supply source between the DC power supply **90** and the battery pack **29a** or **29b** and thus to suppress a cooling failure.

(200) The electric device **1** sends out a refrigerant to the first refrigerant pipe **45** and the second refrigerant pipe **46** (cools the first accommodation chamber **9** and the second accommodation chamber **10**) with the single compressor **41**, but if it is attempted to send out the refrigerant to the first refrigerant pipe **45** and the second refrigerant pipe **46** at the same time (cool the first accommodation chamber **9** and the second accommodation chamber **10** at the same time), temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** may deviate from set temperatures (both chambers cannot reach the set temperatures) due to dimensional errors in diameters of the first refrigerant pipe **45** and the second refrigerant pipe **46**, a length difference caused by a difference in a structure in which the first refrigerant pipe **45** and the second refrigerant pipe **46** are laid, or dimensional errors of the two systems of capillaries connected to the first refrigerant pipe **45** and the second refrigerant pipe **46**.

(201) In order to deal with this problem, the electric device **1** can perform an alternating operation mode. The alternating operation mode is a mode in which alternating cooling is performed by repeatedly causing a refrigerant to flow through only the first refrigerant pipe **45** and the refrigerant to flow through only the second refrigerant pipe **46**. In the alternating operation mode, turning-on (opening) of the first regulating valve **47a** and turning-on (opening) of the second regulating valve **47b** do not overlap.

(202) FIG. **49** is a flowchart illustrating an operation in a two-room mode in which the alternating

operation mode is added to the two-room mode in FIG. 45. In this flowchart, determination 1 (S521) and determination 2 (S522) for determining whether to transition to the alternating operation mode are added after S52 in FIG. 45. In a case where conditions of determination 1 (S521) and determination 2 (S522) are both satisfied (Yes in S521 and Yes in S522), the microcomputer 81 transitions to the alternating operation mode (S523). In a case where at least one of the conditions of determination 1 (S521) and determination 2 (S522) is not satisfied (No in S521 or No in S522), the microcomputer 81 proceeds to S53 in FIG. 45.

(203) FIG. 50 is a diagram for describing a specific example of determination 1 (S521) in FIG. 49. Determination 1 is a first condition necessary for transitioning to the alternating operation mode, and is, for example, one of the following patterns. •Pattern 1: It is determined whether a set temperature of at least one of the first accommodation chamber 9 and the second accommodation chamber 10 is lower than a designated temperature (for example, 0° C.). •Pattern 2: It is determined whether set temperatures of both the first accommodation chamber 9 and the second accommodation chamber 10 are lower than a designated temperature (for example, 0° C.). If a result of determination 1 is Yes, the process proceeds to determination 2 (S522).

(204) FIG. 51 is a diagram for describing a specific example of determination 2 (S522) in FIG. 49. Determination 2 is a second condition necessary for transitioning to the alternating operation mode, and is, for example, one of the following patterns. •Pattern 1: It is determined whether a temperature of at least one of the first accommodation chamber 9 and the second accommodation chamber 10 is lower than a designated temperature (for example, 0° C.). •Pattern 2: It is determined whether temperatures of both the first accommodation chamber 9 and the second accommodation chamber 10 are lower than a designated temperature (for example, 0° C.). •Pattern 3: It is determined whether simultaneous cooling in which a refrigerant is sent out to the first refrigerant pipe and the second refrigerant pipe 46 at the same time (cooling the first accommodation chamber 9 and the second accommodation chamber 10 at the same time) has been executed for a designated time (for example, 20 minutes). •Pattern 4: It is determined whether the outside air temperature exceeds a designated temperature (for example, 35° C.). Pattern 5: It is determined whether a drive current for the compressor 41 exceeds a designated current value (8 A, for example). •Pattern 6: It is determined whether an input voltage of the compressor 41 is less than a designated voltage value (12 V). •Pattern 7: It is determined whether a temperature difference between the first accommodation chamber 9 and the second accommodation chamber 10 exceeds a designated temperature difference (for example, 2° C.). In a case where a result of determination 2 is Yes, that is, in a case where the predetermined state is reached, even if both the temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 have not reached the set temperatures, the simultaneous operation mode for simultaneous cooling transitions to the alternating operation mode (S523).

(205) In determinations 1 and 2, the designated temperatures for the set temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 and the designated temperature (predetermined temperature) for the current temperature are set to temperatures that can be reliably reached as temperatures of the first accommodation chamber 9 and the second accommodation chamber 10 even in the simultaneous operation mode. The designated temperature may be determined to be, for example, the outside air temperature -20° C. in relation to the outside air temperature.

(206) FIG. 52 is a diagram illustrating a specific example of determination 3 (S80) in FIG. 53. Determination 3 is determination or setting regarding to which of the first refrigerant pipe 45 and the second refrigerant pipe 46 the refrigerant is sent out first in the alternating operation mode (which of the first accommodation chamber 9 and the second accommodation chamber 10 is cooled first), and is, for example, one of the following patterns. •Pattern 1: A refrigerant is sent out first to a refrigerant pipe corresponding to a higher temperature out of the first accommodation chamber 9 and the second accommodation chamber 10 (cooling is performed from a higher temperature

room). •Pattern 2: A refrigerant is sent out first to the first refrigerant pipe **45** (cooled from the first accommodation chamber **9**). •Pattern 3: A refrigerant is sent out first to the second refrigerant pipe **46** (cooled from the second accommodation chamber **10**). In determination 3, in a case where the refrigerant is sent out first to the first refrigerant pipe (cooled from the first accommodation chamber **9**), **1** is assigned to a flag, and in a case where the refrigerant is sent out first to the second refrigerant pipe **46** (cooling from the second accommodation chamber **10**), **0** is assigned to the flag. (207) FIG. **53** is a flowchart illustrating a first pattern of the alternating operation mode (S523) in FIG. **49**. In the first pattern of the alternating operation mode, in a case where a temperature difference between the first accommodation chamber **9** and the second accommodation chamber **10** exceeds a designated temperature difference (for example, 2° C.), a refrigerant pipe to which the refrigerant is sent out is changed (a room to be cooled). After determination 3 (S81), if the flag is 0 (Yes in S82), the microcomputer **81** reads set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** (S83), and reads output signals from the first thermistor **55** and the second thermistor **56** (the current temperature of the first accommodation chamber **9** and the second accommodation chamber **10**) (S84). In a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature (Yes in S85) and in a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature (Yes in S86), the microcomputer **81** stops the compressor **41** (S87). In a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature—2° C. (Yes in S88), the microcomputer **81** drives the second heating part **52** (S89), and in other cases (No in S88), stops the second heating part **52** (S90). The microcomputer **81** assigns 1 to the flag (S95), and checks whether or not determination 2 is satisfied (S96).

(208) In a case where the temperature of the first accommodation chamber **9** is not lower than the set temperature in S86 (No in S86), the microcomputer **81** assigns 1 to the flag (S95), and checks whether or not determination 2 is satisfied (S96).

(209) In a case where the current temperature of the second accommodation chamber is not lower than the set temperature in S85 (No in S85), if the current temperature of the second accommodation chamber **10** is not lower than the current temperature of the first accommodation chamber **9**—2° C. (No in S91), the microcomputer **81** turns off (closes) the first regulating valve **47a** and turns on (opens) the second regulating valve **47b** (S92), sets the rotation speed of the compressor **41** to 2,000 rpm (S93), and drives the compressor **41** (S94). As a result, the refrigerant is sent out to the second refrigerant pipe **46** (the refrigerant is not sent out to the first refrigerant pipe **45**). After that, the microcomputer **81** checks whether or not determination 2 is satisfied (S96). If the current temperature of the second accommodation chamber **10** is lower than the current temperature of the first accommodation chamber **9**—2° C. in S91 (Yes in S91), the microcomputer **81** assigns 1 to the flag (S95) and checks whether or not determination 2 is satisfied (S96).

(210) If the flag is 1 in S82 (No in S82), the microcomputer **81** reads set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** (S103), and reads output signals from the first thermistor **55** and the second thermistor **56** (the current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**) (S104). In a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature (Yes in S105) and in a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature (Yes in S106), the microcomputer **81** stops the compressor **41** (S107). In a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature—2° C. (Yes in S108), the microcomputer **81** drives the first heating part **51** (S109), and in other cases (No in S108), stops the first heating part **51** (S110). The microcomputer **81** assigns 0 to the flag (S115), and checks whether or not determination 2 is satisfied (S96).

(211) In a case where the temperature of the second accommodation chamber **10** is not lower than the set temperature in S106 (No in S106), the microcomputer **81** assigns 0 to the flag (S115), and

checks whether or not determination 2 is satisfied (S96).

(212) In a case where the current temperature of the first accommodation chamber **9** is not lower than the set temperature in **S105** (No in **S105**), if the current temperature of the first accommodation chamber **9** is not lower than the current temperature of the second accommodation chamber **10**—2° C. (No in **S111**), the microcomputer **81** turns on (opens) the first regulating valve **47a** and turns off (closes) the second regulating valve **47b** (**S112**), sets the rotation speed of the compressor **41** to 2,500 rpm (**S113**), and drives the compressor **41** (**S114**). As a result, the refrigerant is sent out to the first refrigerant pipe **45** (the refrigerant is not sent out to the second refrigerant pipe **46**). After that, the microcomputer **81** checks whether or not determination 2 is satisfied (**S96**). If the current temperature of the first accommodation chamber **9** is lower than the current temperature of the second accommodation chamber **10**—2° C. in **S111** (Yes in **S111**), the microcomputer **81** assigns 0 to the flag (**S115**) and checks whether or not determination 2 is satisfied (**S96**).

(213) In a case where determination 2 is satisfied (Yes in **S96**), the microcomputer **81** returns to **S82**, and if determination 2 is not satisfied (No in **S96**), ends the alternating operation mode and returns to **S51** in FIG. 49.

(214) In FIG. 53, in a case where a cooling target is switched from the second accommodation chamber **10** to the first accommodation chamber **9**, the microcomputer **81** may perform control for switching opening and closing states of the regulating valve **47** and then increasing an output from (rotation speed of) the compressor **41**, and in a case where a cooling target is switched from the first accommodation chamber **9** to the second accommodation chamber **10**, the microcomputer **81** may perform control for decreasing an output from (rotation speed of) the compressor **41** and then switching opening and closing states of the regulating valve **47**.

(215) In FIG. 53, in a case where the set temperature of at least one of the first accommodation chamber **9** and the second accommodation chamber **10** is changed and determination 2 is satisfied to enter the alternating operation mode, the microcomputer **81** may cause a refrigerant to be sent out first to only a refrigerant pipe corresponding to one accommodation chamber with the higher current temperature. After that, the microcomputer **81** may change a refrigerant pipe to which a refrigerant is to be sent out when a slope of the temperature in the other accommodation chamber changes from negative to positive, or when the current temperature is reversed and a temperature difference reaches a predetermined value (for example, 2° C.) or more.

(216) FIG. 54 is a flowchart illustrating a second pattern of the alternating operation mode (**S523**) in FIG. 49. In the second pattern of the alternating operation mode, the refrigerant pipes to which the refrigerant is to be sent out (rooms to be cooled) are switched according to time. In a case of cooling either the first accommodation chamber **9** or the second accommodation chamber **10**, the rotation speed of the compressor **41** is set to 2,000 rpm, and one cooling time for the first accommodation chamber **9** having a large volume is longer than that of the second accommodation chamber **10** having a small volume. After determination 3 (**S81**), the microcomputer **81** starts a timer (**S122**).

(217) In a case where the flag is 0 (Yes in **S123**) and the elapsed time of the timer has not exceeded 300 seconds that is an example of a second predetermined time (No in **S124**), the microcomputer **81** reads set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** (**S125**), and reads output signals from the first thermistor **55** and the second thermistor **56** (the current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**) (**S126**).

(218) In a case where the current temperature of the second accommodation chamber is lower than the set temperature (Yes in **S127**) and in a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature (Yes in **S128**), the microcomputer **81** stops the compressor **41** (**S129**). In a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature—2° C. (Yes in **S130**), the

microcomputer **81** drives the second heating part **52** (S131), and in other cases (No in S130), stops the second heating part **52** (S132). The microcomputer **81** assigns 1 to the flag (S136), resets the timer (S137), and checks whether or not determination 2 is satisfied (S138).

(219) In a case where the temperature of the first accommodation chamber **9** is not lower than the set temperature in S128 (No in S128), the microcomputer **81** assigns 1 to the flag (S136), resets the timer (S137), and checks whether determination 2 is satisfied (S138).

(220) In a case where the current temperature of the second accommodation chamber is not lower than the set temperature in S127 (No in S127), the microcomputer **81** turns off (closes) the first regulating valve **47a** and turns on (opens) the second regulating valve **47b** (S133), sets the rotation speed of the compressor **41** to 2,000 rpm (S134), and drives the compressor **41** (S135). As a result, the refrigerant is sent out to the second refrigerant pipe **46** (the refrigerant is not sent out to the first refrigerant pipe **45**). After that, the microcomputer **81** checks whether or not determination 2 is satisfied (S138).

(221) In a case where the elapsed time of the timer exceeds 300 seconds in S124 (Yes in S124), the microcomputer **81** assigns 1 to the flag (S136), resets the timer (S137), and checks whether determination 2 is satisfied (S138).

(222) In a case where the flag is 1 in S123 (Yes in S123) and the elapsed time of the timer has not exceeded 450 seconds that is an example of a first predetermined time (No in S144), the microcomputer **81** reads set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** (S145), and reads output signals from the first thermistor **55** and the second thermistor **56** (the current temperatures of the first accommodation chamber **9** and the second accommodation chamber **10**) (S146).

(223) In a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature (Yes in S147) and in a case where the current temperature of the second accommodation chamber **10** is lower than the set temperature (Yes in S148), the microcomputer **81** stops the compressor **41** (S149). In a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature—2° C. (Yes in S150), the microcomputer **81** drives the first heating part **51** (S151), and in other cases (No in S150), stops the first heating part **51** (S152). The microcomputer **81** assigns 0 to the flag (S156), resets the timer (S157), and checks whether or not determination 2 is satisfied (S138).

(224) In a case where the temperature of the second accommodation chamber **10** is not lower than the set temperature in S148 (No in S148), the microcomputer **81** assigns 0 to the flag (S156), resets the timer (S157), and checks whether determination 2 is satisfied (S138).

(225) In a case where the current temperature of the first accommodation chamber **9** is not lower than the set temperature in S147 (No in S147), the microcomputer **81** turns on (opens) the first regulating valve **47a** and turns off (closes) the second regulating valve **47b** (S153), sets the rotation speed of the compressor **41** to 2,000 rpm (S154), and drives the compressor **41** (S155). As a result, the refrigerant is sent output to the first refrigerant pipe **45** (the refrigerant is not sent out to the second refrigerant pipe **46**). After that, the microcomputer **81** checks whether or not determination 2 is satisfied (S138).

(226) In a case where the elapsed time of the timer exceeds 450 seconds in S144 (Yes in S144), the microcomputer **81** assigns 0 to the flag (S156), resets the timer (S157), and checks whether determination 2 is satisfied (S138).

(227) In a case where determination 2 is satisfied (Yes in S138), the microcomputer **81** returns to S123, and in a case where determination 2 is no longer satisfied (No in S138), ends the alternating operation mode and returns to S51 in FIG. 49.

(228) In the second pattern of the alternating operation mode, the rotation speed of the compressor **41** in a case of cooling the first accommodation chamber **9** may be set to 2,500 rpm, the rotation speed of the compressor **41** in a case of cooling the second accommodation chamber **10** may be set to 2,000 rpm, and one cooling time may be the same in a case of cooling either the first

accommodation chamber **9** or the second accommodation chamber **10**. The pattern changed as described above is a third pattern of the alternating operation mode.

(229) FIG. **55** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where set temperatures of both the first accommodation chamber **9** and the second accommodation chamber **10** are -18°C ., and an operation associated with transition from the simultaneous operation mode to the third pattern of the alternating operation mode is performed in the two-room mode. At 0 minutes, the temperatures of both the first accommodation chamber **9** and the second accommodation chamber **10** are 30°C . The microcomputer **81** is in the simultaneous operation mode in a period from 0 minutes to 20 minutes, sets the rotation speed of the compressor **41** to 3,000 rpm, and simultaneously cools the first accommodation chamber **9** and the second accommodation chamber **10**. The microcomputer **81** transitions to the alternating operation mode when the temperature of the second accommodation chamber that is a small room becomes lower than 0°C . at 20 minutes. In the alternating operation mode, the microcomputer **81** starts cooling the first accommodation chamber **9** (large room) of which the temperature is high, and then switches cooling target rooms every 300 seconds. As a result, the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** approach the set temperature of -18°C . The microcomputer **81** sets the rotation speed of the compressor **41** to 2,500 rpm when cooling only the first accommodation chamber **9** and to 2,000 rpm when cooling only the second accommodation chamber **10**.

(230) In FIG. **55**, the cooling of the second accommodation chamber **10** is stopped at 20 minutes, but the temperature drop of the second accommodation chamber **10** continues until about 22 minutes. This is due to heat exchange between the low-temperature refrigerant remaining in the second refrigerant pipe **46** and the second accommodation chamber **10**. Similarly, even in a case where the cooling of the first accommodation chamber **9** is stopped, the temperature drop of the first accommodation chamber **9** continues for several minutes. As described above, even if the cooling of one room is stopped, the temperature drop of the one room continues for several minutes after the cooling is stopped. The same applies to FIGS. **56** and **57** as well.

(231) FIG. **56** is a graph illustrating temporal changes in temperature of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where set temperatures of both the first accommodation chamber **9** and the second accommodation chamber **10** are -18°C ., and an operation associated with transition from the simultaneous operation mode to the second pattern of the alternating operation mode is performed in the two-room mode. At 0 minutes, the temperatures of both the first accommodation chamber **9** and the second accommodation chamber **10** are 25°C . The microcomputer **81** is in the simultaneous operation mode in a period from 0 minutes to 15 minutes, sets the rotation speed of the compressor **41** to 3,000 rpm, and simultaneously cools the first accommodation chamber **9** and the second accommodation chamber **10**. The microcomputer **81** transitions to the alternating operation mode when the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** become lower than 0°C . at 15 minutes. In the alternating operation mode, the microcomputer **81** starts cooling the second accommodation chamber **10**, sets one cooling time to 300 seconds for the second accommodation chamber **10** and to 450 seconds for the first accommodation chamber **9**, and switches cooling target rooms. As a result, the temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** approach the set temperature of -18°C . The microcomputer **81** sets the rotation speed of the compressor **41** to 2,000 rpm both when cooling only the first accommodation chamber **9** and when cooling only the second accommodation chamber **10**.

(232) FIG. **57** is a graph illustrating temporal changes in temperature of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is -18°C ., a set temperature

of the second accommodation chamber **10** is -5°C ., and an operation associated with transition from the simultaneous operation mode to further another pattern of the alternating operation mode is performed in the two-room mode. At 0 minutes, the temperatures of both the first accommodation chamber **9** and the second accommodation chamber **10** are 25°C . The microcomputer **81** is in the simultaneous operation mode in a period from 0 minutes to 20 minutes, sets the rotation speed of the compressor **41** to 3,000 rpm, and simultaneously cools the first accommodation chamber **9** and the second accommodation chamber **10**. The microcomputer **81** transitions to the alternating operation mode when the temperature of the second accommodation chamber **10** that is a small room becomes lower than the set temperature (-5°C .) at 20 minutes. In the alternating operation mode, the microcomputer **81** first cools only the first accommodation chamber **9**, and if the temperature of the second accommodation chamber **10** exceeds the set temperature $+2^{\circ}\text{C}$., switches to cooling only the second accommodation chamber **10**. If the temperature of the second accommodation chamber **10** becomes lower than the set temperature, the microcomputer **81** switches to cooling only the first accommodation chamber **9**. If the temperature of the first accommodation chamber **9** reaches the set temperature while only the first accommodation chamber **9** is being cooled, the microcomputer **81** stops cooling the first accommodation chamber **9**, and if the temperature of the second accommodation chamber **10** does not exceed the set temperature $+2^{\circ}\text{C}$., does not cool the second accommodation chamber **10** (stops the compressor **41**). If the temperature of the second accommodation chamber **10** exceeds the set temperature $+2^{\circ}\text{C}$., the microcomputer **81** cools only the second accommodation chamber **10**. The microcomputer **81** sets the rotation speed of the compressor **41** to 2,500 rpm when cooling only the first accommodation chamber **9** and to 2,000 rpm when cooling only the second accommodation chamber **10**.

(233) By executing the alternating operation mode as described above, the electric device **1** can make temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** reliably reach set temperatures and control temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** with high accuracy even if there are dimensional errors in diameters of the first refrigerant pipe **45** and the second refrigerant pipe **46**, a length difference caused by a difference in a structure in which the first refrigerant pipe **45** and the second refrigerant pipe **46** are laid, or dimensional errors of the two systems of capillaries connected to the first refrigerant pipe and the second refrigerant pipe **46**.

(234) FIG. **58** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is 60°C ., a set temperature of the second accommodation chamber **10** is 30°C ., and cooling of the second accommodation chamber **10** using the compressor **41** is stopped by controlling a temperature of the second accommodation chamber **10** in the two-room mode.

(235) FIG. **59** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is 60°C ., a set temperature of the second accommodation chamber **10** is 30°C ., and cooling of the second accommodation chamber **10** using the compressor **41** is stopped by controlling a time from the start of the cooling in the two-room mode.

(236) In both FIGS. **58** and **59**, the microcomputer **81** controls to keep the temperature of the second accommodation chamber **10** at 30°C . by cooling the second accommodation chamber **10** when the temperature of the second accommodation chamber **10** is approaching 60°C . over a set temperature of 30°C . under the influence of the temperature of the first accommodation chamber **9** being 60°C .

(237) In the case of FIG. **58**, in a case where the temperature of the second accommodation chamber **10** exceeds 32°C . (30°C . $+2^{\circ}\text{C}$.), cooling of only the second accommodation chamber

10 is started, and if the temperature of the second accommodation chamber **10** reaches 28° C. (30° C.–2° C.) while only the second accommodation chamber **10** is being cooled, the cooling of the second accommodation chamber **10** is stopped (the compressor **41** is stopped). However, even if cooling of the second accommodation chamber **10** is stopped as described above, since the temperature of the second accommodation chamber **10** continues to drop for several minutes, the second heating part **52** is driven due to the temperature drop of the second accommodation chamber **10** with respect to the set temperature. Therefore, cooling and heating of the second accommodation chamber **10** are repeated, which is not preferable in terms of power consumption. (238) In the case of FIG. **59**, in a case where the temperature of the second accommodation chamber **10** exceeds 32° C. (30° C. +2° C.), cooling of only the second accommodation chamber **10** is started and stopped continuously for one minute (predetermined time). By restricting the time for cooling only the second accommodation chamber **10**, it is possible to prevent the second accommodation chamber **10** from being too cold with respect to the set temperature, to reduce the driving of the second heating part **52**, and thus to reduce power consumption.

(239) FIG. **60** is a flowchart illustrating an operation in a two-room mode in which time control is added to the two-room mode in FIG. **45**. The time control is to intermittently operate the compressor **41** in a case where the number of cooling target room is one and a set temperature of the room exceeds a predetermined temperature. Hereinafter, processes added to FIG. **45** will be mainly described. In FIG. **60**, 20° C. is an example of normal temperature. An outside air temperature may be used instead of 20° C.

(240) In a case where the current temperature of the first accommodation chamber **9** is higher than the set temperature+2° C. (Yes in S53) and the current temperature of the second accommodation chamber **10** is not higher than the set temperature+2° C. (No in S54), if the set temperature of the first accommodation chamber **9** is higher than 20° C. (Yes in S541), the microcomputer **81** sets the rotation speed of the compressor **41** to 2,500 rpm (S542), turns on (opens) the first regulating valve **47a** and turns off (closes) the second regulating valve **47b** (S543), drives the compressor **41** for a designated time (S544), stops the compressor **41** for a designated time (S545), and returns to S51. In a case where the set temperature of the first accommodation chamber **9** is equal to or lower than 20° C. in S541 (No in S541), the microcomputer **81** proceeds to S58 in FIG. **45**.

(241) In a case where the current temperature of the first accommodation chamber **9** is lower than the set temperature—2° C. (Yes in S61) and the current temperature of the second accommodation chamber **10** is higher than the set temperature+2° C. (Yes in S62), if the set temperature of the second accommodation chamber **10** is higher than 20° C. (Yes in S621), the microcomputer **81** sets the rotation speed of the compressor **41** to 2,000 rpm (S622), turns off (closes) the first regulating valve **47a** and turns on (opens) the second regulating valve **47b** (S623), drives the compressor **41** for a designated time (S624), drives the first heating part **51** and stops the second heating part **52** (S625), stops the compressor **41** for a designated time (S626), and returns to S51. In a case where the set temperature of the second accommodation chamber **10** is equal to or lower than 20° C. in S621 (No in S621), the microcomputer **81** proceeds to S63 in FIG. **45**.

(242) In a case where the current temperature of the first accommodation chamber **9** is not higher than the set temperature+2° C. and not lower than the set temperature—2° C. (No in S53 and No in S61), and the current temperature of the second accommodation chamber is higher than 20° C. (Yes in S701), if the set temperature of the second accommodation chamber **10** is higher than 20° C. (Yes in S701), the microcomputer **81** sets the rotation speed of the compressor **41** to 2,000 rpm (S702), turns off (closes) the first regulating valve **47a** and turns on (opens) the second regulating valve **47b** (S703), drives the compressor **41** for a designated time (S704), stops the compressor **41** for a designated time (S705), and returns to S51. In a case where the set temperature of the second accommodation chamber **10** is equal to or lower than 20° C. in S701 (No in S701), the microcomputer **81** proceeds to S71 in FIG. **45**.

(243) (A) of FIG. **61** is a waveform diagram of a drive current for the compressor **41** before and

after an operation is resumed in a case where an operation of the compressor **41** is performed, stopped for one minute, and then resumed. (B) of FIG. **61** is a waveform diagram of a drive current for the compressor **41** before and after an operation is resumed in a case where an operation of the compressor **41** is performed, stopped for two minutes, and then resumed. As illustrated in (A) of FIG. **61**, in a case where the compressor **41** is stopped for one minute, a starting current for the compressor **41** at the time of resuming an operation is as large as 12.2 A. In contrast, as illustrated in (B) of FIG. **61**, in a case where the stop time of the compressor **41** is two minutes, the starting current for the compressor **41** is relatively suppressed to 5.2 A when the operation is resumed. Therefore, it is desirable that a designated time for stopping the compressor **41** in **S545**, **S626**, and **S705** in FIG. **60** is two minutes or longer.

(244) FIG. **62** is a graph illustrating temporal changes in temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** and a rotation speed of the compressor **41** in a case where a set temperature of the first accommodation chamber **9** is 60° C., a set temperature of the second accommodation chamber **10** is 30° C., and cooling of the second accommodation chamber **10** using the compressor **41** is performed by repeatedly performing an operation for one minute and stoppage for two minutes in the two-room mode. At 0 minutes, temperatures of both the first accommodation chamber **9** and the second accommodation chamber **10** are 60° C. From here, the microcomputer **81** controls to decrease the temperature of the second accommodation chamber **10** to 30° C. by cooling the second accommodation chamber **10** when the temperature of the second accommodation chamber **10** is difficult to decrease due to the influence of the temperature of the first accommodation chamber **9** being 60° C. In this case, by repeating an operation for one minute and stoppage for two minutes, a starting current for the compressor **41** can be suppressed, and the temperature drop of the second accommodation chamber **10** can be slowed down to prevent the second accommodation chamber **10** from being too cold with respect to the set temperature.

Therefore, it is possible to reduce the driving of the second heating part **52** and reduce power consumption. After the temperature of the second accommodation chamber **10** reaches the set temperature, as in FIG. **59**, control to start cooling only the second accommodation chamber **10** if the temperature of the second accommodation chamber **10** exceeds 32° C. (30° C. +2° C.) and continuously stop the second accommodation chamber **10** for one minute is repeatedly performed.

(245) As illustrated in FIGS. **63** to **66**, the electric device **1** has a storage unit (additional part) **200** that can be attached to and detached from the right side surface. The main frame **11** has a groove part **36** as a first engaged part on the right side part. The right outer box **13** has a pair of projection parts (protrusions) **37** as second engaged parts on both front and rear ends of the lower right side surface. The groove part **36** and the projection parts **37** are disposed at positions spaced apart from each other, and constitute a locking part for locking the storage unit **200**.

(246) As illustrated in FIGS. **68** to **70**, the storage unit **200** includes a bag **201**, a belt **221**, a hook part **222** as a first engaging part, a pair of hook-and-loop fasteners **224**, and a pair of ring parts **226** as a second engaging part, and four buttons **230**. The belt **221** is, for example, three nylon belts (nylon bands) sewn together to form an H shape. The hook part **222** is made of resin, for example, and provided to bridge a pair of upper ends of the belt **221**. Specifically, the upper end of the belt **221** is passed through a through-hole of the hook part **222**, folded back, and sewn. The hook-and-loop fasteners **224** are provided (sewn) on the left and right parts of the belt **221**, respectively. The ring parts **226** are elastic rings (rubber rings), for example, and are provided at a pair of lower ends of the belt **221**, respectively. Specifically, the lower end of the belt **221** is passed through the ring part **226**, folded back, and sewn. The four buttons **230** are provided on the belt **221** and are located above and below each of the pair of hook-and-loop fasteners **224**.

(247) As illustrated in FIG. **70**, the bag **201** has a storage part **202**, a grip part **203**, hook-and-loop fasteners **204** to **206**, cloth parts **208** and **209**, and four buttons **210**. The storage part **202** is made of cloth, for example, has a fastener-type opening and closing mechanism, and has a size capable of storing the partition plate **70** as illustrated in (A) of FIG. **67**. The storage part **201** may be replaced

with a larger storage part **201A** as illustrated in (D) of FIG. **67**. As illustrated in FIG. **70**, the grip part **203** for carrying is made of the same material as the belt **221**, for example, and is provided on the upper surface of the storage part **202**. A pair of hook-and-loop fasteners **204** are provided (sewn) on the back surface of the storage part **202**. The four buttons **210** are provided on the back surface of the storage part **202** and are located above and below each of the pair of hook-and-loop fasteners **204**. The hook-and-loop fastener **205** and the cloth part **208** are provided (sewn) on the rear surface of the storage part **202**. The cloth part **208** continues below the hook-and-loop fastener **205**. The cloth part **209** continues below the cloth part **208**. The hook-and-loop fastener **206** continues below the cloth part **209**.

(248) In a case where the bag **201** is attached to the belt **221**, the pair of hook-and-loop fasteners **204** of the bag **201** and the pair of hook-and-loop fasteners **224** provided on the belt **221** are brought into surface contact with each other and detachably coupled (adhered) to each other. The four buttons **210** of the bag **201** and the four buttons **230** provided on the belt **221** are detachably coupled to each other. The laterally extending portion of the belt **221** is sandwiched between the cloth parts **208** and **209** and folded back at a boundary line between the cloth parts **208** and **209**, so that the hook-and-loop fasteners **205** and **206** are brought into surface contact with each other and are detachably coupled (adhered) to each other.

(249) In a case where the storage unit **200** is attached to the electric device **1**, the hook part **222** is engaged (hooked) with the groove part **36** and the ring part **226** is engaged (hooked) with the projection part **37** as illustrated in FIG. **63** in a state in which the second lid **7** is open as illustrated in FIG. **4**. A separation distance between the groove part **36** and the projection part **37**, a vertical length of the belt **221**, and a diameter of the ring part **226** have a dimensional relationship in which the ring part **226** is hooked on the projection part **37** in a state of being stretched beyond its natural length. Thus, in a case where the storage unit **200** is attached to the electric device **1**, tension (tensile force) is applied to the belt **221** in the up-down direction due to the elastic force of the ring part **226**. Consequently, the bag **201** is suppressed from wobbling and the bag **201** is stably held.

(250) In a state in which the groove part **36** and the hook part **222** are engaged with each other, the second lid **7** in a closed state prevents the hook part **222** from coming off. This is because, as illustrated in (A) of FIG. **66**, a distance $L1$ from the upper end of the hook part **222** to the portion of the second lid **7** facing the upper end in a in synchronization with the groove part **36** and the hook part **222** engaged is less than a thickness (vertical length) $L2$ of the hook part **222** ($L1 < L2$). Since the second lid **7** prevents the hook part **222** from coming off, it is possible to suppress separation of the storage unit **200** from the electric device **1** due to vibration or the like.

(251) Since the storage unit **200** is attachable to and detachable from the electric device **1**, it is possible to add an accommodation function other than the heat insulation or cold insulation space (the first accommodation chamber **9** and the second accommodation chamber **10**), and thus convenience is high. By attaching the storage unit **200** to the electric device **1**, small items such as a battery pack and a bottle opener can be stored in place of the bag **201**, and thus it is convenient to carry small items that do not require heating or cooling. The detached partition plate **70** can be stored in the bag **201**, and thus it is convenient to carry and place the detached partition plate **70**.

(252) As illustrated in FIG. **71**, in addition to the storage unit **200**, accessory cases **231** to **233** and an S-shaped hook **237** are attachable to and detachable from the electric device **1**. The second lid **7** has a hooking part (groove part) **91** capable of hooking (locking) the accessory case **231**, a hooking part **102** (groove part) capable of hooking (locking) the accessory case **233**, and hooking parts (through holes) **93** and **94** capable of hooking (locking) the S-shaped hook **237**. The first lid **6** has a hooking part (groove part) **92** capable of hooking (locking) the accessory case **232**, and hooking parts (through holes) **95** and **96** capable of hooking (locking) the S-shaped hook **237**. The accessory cases **231** to **233** are convenient as temporary storerooms. The S-shaped hook **237** is convenient for hooking small items. Although the accessory case **231** can be hooked on the groove part **36** as illustrated in FIG. **72**, the storage unit **200** cannot be attached in this case.

(253) As illustrated in FIGS. 73 and 74, in the electric device 1, a storage unit (additional part) 250 is attachable to and detachable from the movable handle 20. The movable handle 20 has a grip 20a and a bridge part 20b. The storage unit 250 has a bag 251 and two belts 253. Each belt 253 has one end sewn to the bag 251 and extends to surround the bridge part 20b between the belt and the rear surface of the bag 251, and a hook-and-loop fastener 254 provided (sewn) on the other end side of the bag 251 is brought into surface contact with a hook-and-loop fastener (not illustrated) provided (sewn) on the rear surface of the bag 251 and detachably coupled (adhered) to each other. Button-to-button coupling may be used instead of or in addition to hook-and-loop fastener. A finger insertion space 20c is secured between the bag 251 and the grip 20a, and the use of the movable handle 20 is not hindered. The storage unit 250 also achieve the same effects as the storage unit 200 (addition of the accommodation function). The electric device 1 may be equipped with both storage units 200 and 250.

(254) FIGS. 75 to 81 relate to an electric device 1A according to another embodiment of the present invention. The electric device 1A differs from the electric device 1 in that storage units 300, 400, and 500 as additional parts are attachable and detachable. The following description will focus on the differences.

(255) As illustrated in FIGS. 75 and 76, the electric device 1 has the storage unit (additional part) 300 attachable to and detachable from its front surface. The main frame 11 has a pair of groove parts 38 as first engaged parts at both left and right ends of the front side part. The left outer box 12 has a pair of projection parts (protrusions) 39 as second engaged parts on both left and right ends of the lower front surface. The groove part 38 and the projection parts 39 are disposed at positions spaced apart from each other, and constitute a locking part for locking the storage unit 300.

(256) The storage unit 300 has a bag 301, a belt 321, a pair of hook parts 322 as first engaging parts, a pair of hook-and-loop fasteners 324, a pair of ring parts 326 as second engaging parts, and four buttons 330. The belt 321 is, for example, three nylon belts (nylon bands) sewn together to form an H shape. The pair of hook parts 322 are made of resin, for example, and are provided at a pair of upper ends of the belt 321, respectively. Specifically, the upper end of the belt 321 is passed through a through-hole of the hook part 322, folded back, and sewn. The hook-and-loop fasteners 324 are provided (sewn) on the left and right parts of the belt 321, respectively. The ring parts 326 are elastic rings (rubber rings), for example, and are provided at a pair of lower ends of the belt 321, respectively. Specifically, the lower end of the belt 321 is passed through the ring part 326, folded back, and sewn. The four buttons 330 are provided on the belt 321 and are located above and below each of the pair of hook-and-loop fasteners 324.

(257) The bag 301 has the same configuration as that of the bag 201 except that the bag 301 does not have a grip part. However, the bag 301 may have a handle. Attachment of the bag 301 to the belt 321 may be performed in the same manner as attachment of the bag 201 to the belt 221.

(258) In a case where the storage unit 300 is attached to the electric device 1A, the hook part 322 is engaged (hooked) with the groove part 38 and the ring part 326 is engaged (hooked) with the projection part 39 as illustrated in FIG. 76 in a state in which the first lid 6 is open as illustrated in FIG. 2. A separation distance between the groove part 38 and the projection part 39, a vertical length of the belt 321, and a diameter of the ring part 326 have a dimensional relationship in which the ring part 326 is hooked on the projection part 39 in a state of being stretched beyond its natural length. Thus, in a case where the storage unit 300 is attached to the electric device 1A, tension (tensile force) is applied to the belt 321 in the up-down direction due to the elastic force of the ring part 326. Consequently, the bag 301 is suppressed from wobbling and the bag 301 is stably held.

(259) In a state in which the groove part 38 and the hook part 322 are engaged with each other, the first lid 6 in a closed state prevents the hook part 322 from coming off. This is for the same reason that the second lid 7 in a closed state prevents the hook part 222 from coming off in a state in which the groove part 36 is engaged with the hook part 222 as described above. Since the first lid 6 prevents the hook part 322 from coming off, it is possible to suppress separation of the storage unit

300 from the electric device **1A** due to vibration or the like.

(260) Since the storage unit **300** is attachable to and detachable from the electric device **1**, it is possible to add an accommodation function other than the heat insulation or cold insulation space (the first accommodation chamber **9** and the second accommodation chamber **10**), and thus convenience is high. By attaching the storage unit **300** to the electric device **1A**, small items such as a battery pack and a bottle opener can be stored in place of the bag **301**, and thus it is convenient to carry small items that do not require heating or cooling. The removed partition plate **70** can be stored in the bag **301**, and thus it is convenient to carry and place the detached partition plate **70**.

(261) As illustrated in FIGS. **77** and **78**, the electric device **1A** has the storage unit **200** attachable to and detachable from the right surface, the storage unit **300** attachable to and detachable from the front surface, and the storage unit **400** attachable to and detachable from the rear surface (back surface). Attachment of the storage units **200** and **300** is as described above. The storage unit **400** has the same configuration as that the storage unit **300**, and can be attached to the rear surface of the electric device **1A** in the same manner as the storage unit **300** attached to the front surface of the electric device **1A**. The main frame **11** also has a pair of groove parts **38** as first engaged parts on the rear side part. The groove part **38** is engaged with a hook part (first engaging part) **422** of the storage unit **400**. The left outer box **12** also has a pair of projection parts (protrusions) **39** as second engaged parts on both left and right ends of the lower rear surface. The projection part **39** is engaged with a ring part (second engaging part) **426** of the storage unit **400**.

(262) The belt **221** of the storage unit **200** and the belt **321** of the storage unit **300** are connected to each other via a connection part **240**. The belt **221** of the storage unit **200** and the belt **421** of the storage unit **400** are connected to each other via a connection part **241**. That is, the storage units **200**, **300**, and **400** are connected to each other via the connection parts **240** and **241**. However, the connection parts **240** and **241** may be omitted, and the storage units **200**, **300** and **400** may be separated from each other.

(263) As illustrated in FIGS. **79** to **81**, the electric device **1A** has a storage unit **500** attachable to and detachable from its upper surface. The first lid **6** has a pair of downward facing groove parts **97** as first engaged parts at both left and right ends of the front side part, and downward facing groove parts **98** as second engaged parts at the center of the rear side part. The groove parts **97** and **98** are disposed at positions spaced apart from each other and constitute locking parts for locking the storage unit **500**.

(264) The storage unit **500** has a bag **501**, a belt **521**, hook parts **522**, a hook-and-loop fastener **524**, and buttons **530**. The hook parts **522** are respectively provided at the front end of the bifurcated belt **521** and the rear end of the belt **521**. The hook part **522** on the front side is hooked on the groove part **97**. The hook part **522** on the rear side is hooked on the groove part **98**. The hook-and-loop fastener **524** and the buttons **530** are provided on the belt **521** for attachment of the bag **501**. The bag **501** has hook-and-loop fasteners and buttons (not illustrated) corresponding to the hook-and-loop fastener **524** and buttons **530** on the rear surface. The storage unit **500** also achieve the same effects as the storage unit **200** (addition of the accommodation function).

(265) The second lid **7** has a downward facing groove part **99** as a first engaged part on the front side part, and a pair of downward facing groove parts **100** as second engaged parts at both left and right ends of the rear side part. Although not illustrated, a storage unit having a configuration similar to the storage unit **500** is attachable to and detachable from the electric device **1A** by using the groove parts **99** and **100**.

(266) FIG. **82** is a perspective view of an electric device **1B** according to still another embodiment of the present invention. The following description will focus on differences from the electric device **1**. The electric device **1B** has groove parts **101** as first engaged parts at both left and right ends of the front side part of the first lid **6**. The left outer box **12** has a pair of projection parts (protrusions) **39** as second engaged parts at both left and right ends of the lower front surface.

(267) A belt **621** is, for example, three nylon belts (nylon bands) sewn together to form an H shape.

A pair of hook parts **622** are made of resin, for example, and are provided at the pair of upper ends of the belt **621**, respectively. Specifically, the upper end of the belt **621** is passed through a through-hole of the hook part **622**, folded back, and sewn. Hook-and-loop fasteners **624** are provided (sewn) on the left and right parts of the belt **621**, respectively. Ring parts **626** are elastic rings (rubber rings), for example, and are provided at a pair of lower ends of the belt **621**, respectively. Specifically, the lower end of the belt **621** is passed through the ring part **626**, folded back, and sewn. Four buttons **630** are provided on the belt **621** and located at the top and bottom of the pair of hook-and-loop fasteners **624**. A bag (not illustrated) having the same configuration as that of the bag **301** in FIG. **75** can be attached to the belt **621**, and can constitute a storage unit.

(268) In a case where this storage unit is attached to the electric device **1B**, the hook part **622** is engaged (hooked) with the groove part **101** and the ring part **626** is engaged (hooked) with the projection part **39** in a state in which the first lid **6** is closed. A separation distance between the groove part **101** and the projection part **39**, a vertical length of the belt **621**, and a diameter of the ring part **626** have a dimensional relationship in which the ring part **626** is hooked on the projection part **39** in a state of being stretched beyond its natural length, and thus tension (tensile force) is applied to the belt **621** in the up-down direction due to the elastic force of the ring part **626**. As a result, a bag (not illustrated) attached to the belt **621** is suppressed from wobbling, and the bag is stably held. The belt **621** prevents the first lid **6** from being unintentionally opened due to overturning, vibration, or the like of the electric device **1B**.

(269) FIG. **83** is a perspective view of the inside of the electric device **1** in a state in which the accommodation chamber **8** is divided into three by using the partition plates **70a** and **70b** (three-room mode), viewed from the front upper right. Here, a flow path of a refrigerant will be described. A structure and control other than the flow path of the refrigerant are basically as described above. A refrigerant flowing out of the compressor **41** flows into the regulating valve **703** via the condenser **42**. The regulating valve **703** is connected to the first refrigerant pipe **45** as a first cooling part, the second refrigerant pipe **46** as a second cooling part, and the fifth refrigerant pipe **701** as a fifth cooling part. The refrigerant that has flowed into the regulating valve **703** flows out to three flow paths at most. That is, the refrigerant that has flowed into the regulating valve **703** through one flow path is divided into a maximum of three flow paths (the first refrigerant pipe **45**, the second refrigerant pipe **46**, and the fifth refrigerant pipe **701**) and flows out. The regulating valve **703** includes a third regulating valve capable of controlling the flow of the first refrigerant pipe **45**, a fourth regulating valve capable of controlling a flow in the second refrigerant pipe **46**, and a fifth regulating valve capable of controlling a flow in the fifth refrigerant pipe **701** inside thereof. Each of these regulating valves can switch between an open state and a closed state in response to a signal from the microcomputer **81**, and can control whether or not the refrigerant flows through each refrigerant pipe by switching between the open state and the closed state. Specific details of the control are as described above.

(270) The first refrigerant pipe **45** is provided on at least the side surface of the first accommodation chamber **9** as described above, and cools the first accommodation chamber **9**. The second refrigerant pipe **46** is provided on at least the side surface of the second accommodation chamber **10** as described above, and cools the second accommodation chamber **10**. The fifth refrigerant pipe **701** is provided on the side surface and the bottom surface of the third accommodation chamber **704** to cool the third accommodation chamber **704**. The first refrigerant pipe **45**, the second refrigerant pipe **46**, and the fifth refrigerant pipe **701** are independent of each other. That is, the first refrigerant pipe **45** is provided to mainly cool the first accommodation chamber **9**, the second refrigerant pipe **46** is provided to mainly cool the second accommodation chamber **10**, and the fifth refrigerant pipe **701** is provided to cool the third accommodation chamber **704**.

(271) Since the first refrigerant pipe **45** and the second refrigerant pipe **46** extend in the same manner as in FIG. **13**, a description thereof will be omitted. The fifth refrigerant pipe **701** extends

along a rear-side member **706** from the upper end to the lower end of the rear surface of the rear-side member **706**. The fifth refrigerant pipe **701** extends toward a front-side member **705** while being folded back on a bottom-side member (not illustrated) left and right after repeating reciprocation of being folded back on the rear side surface of the rear-side member **706** left and right six times. The fifth refrigerant pipe **701** that reaches the front-side member **705** extends from the lower end toward the upper end while being folded back left and right on the front-side member **705**. The fifth refrigerant pipe **701** extending from the lower end to the upper end on the front-side member **705** is connected to the regulating valve **702**. The regulating valve **702** is connected such that the first refrigerant pipe **45**, the second refrigerant pipe **46** and the fifth refrigerant pipe **703** are merged. The refrigerant merged at the regulating valve **702** is discharged from one refrigerant pipe **44** and flows into the compressor **41**. As described above, by disposing the three refrigerant pipes to extend independently to the three accommodation chambers, it is possible to independently control each of the accommodation chambers in a case where it is desired to use the three-room mode.

(272) Although the present invention has been described above by using the embodiments as examples, it will be understood by those skilled in the art that various modifications can be made to each constituent and each processing process of the embodiments within the scope of the claims. Modification examples will be described below.

(273) The first accommodation chamber **9** and the second accommodation chamber **10** may have the same size. The first refrigerant pipe **45** and the second refrigerant pipe **46** may be separate from the right-side member **16** and the left-side member **17**. Instead of a configuration in which, in the setting part **60**, by pressing the execution button **67**, set temperatures of the first accommodation chamber **9** and the second accommodation chamber **10** are determined and an operation is started, a configuration in which set temperatures are determined and an operation is started even if the execution button **67** is not pressed may be employed. For example, a configuration may be employed in which set temperatures are determined and an operation is started in a case where three seconds (predetermined time) have elapsed since the right room temperature setting button **62** or the left room temperature setting button **63** was last pressed. In this case, the execution button **67** may be omitted.

(274) Materials of the belt, the bag, the hook part, the ring part, and the like in the storage unit may be changed as appropriate. In the embodiments of attaching and detaching the storage unit, instead of the cooling mechanism **40** using the compressor **41**, another cooling mechanism such as a Peltier element may be provided.

(275) A connectable number of battery packs, a rated output voltage of a battery pack, a voltage of the DC power supply **90**, various times, various threshold values, and the like, which are exemplified as specific numerical values in the embodiments, do not limit the scope of the invention, and may be changed freely according to required specifications.

REFERENCE SIGNS LIST

(276) **1**, **1A**, **1B** Electric device **2** Main body **3** First body part **4** Second body part **5** Lid **6** First lid **6a** Grip part **7** Second lid **7a** Grip part **8** Accommodation chamber (accommodation part) **8a** Bottom surface **9** First accommodation chamber (first accommodation part) **9a** First side surface **9b** Second side surface **10** Second accommodation chamber (Second accommodation part) **10a** First side surface **10b** Second side surface **11** Main frame **11b** Boss **11c**, **11d** Opening **12** Left outer box **12a** Boss **13** Right outer box **15** Bottom member **16** Right-side member **16a** Outwardly bent part **17** Left-side member **17a** Outwardly bent part **18** Rail member **18a** Notch **18b** Recessed part **18c**, **18d** Groove part **19** Caster **20** Movable handle (carry handle) **20a** Grip **20b** Bridge part **20c** Finger insertion space **21** grip part **22** Battery pack accommodation chamber **22a** Battery pack attachment part **23** Intake port **24** Exhaust port **25** First hinge mechanism **26** Second hinge mechanism **27** USB terminal **28** Power input terminal (in-vehicle power supply connection part) **29** Battery pack **30** Battery box **30a** Boss **30b** Drain hole **31** Battery terminal **32** Branch part **33** First air passage **34**

Second air passage 35 Leg 36 Groove part (first engaged part) 37 Projection part (second engaged part) 38 Groove part (first engaged part) 39 Projection part (second engaged part) 40 Cooling mechanism 41 Compressor (cooler) 42 Condenser 43 Capillary tube 44 Refrigerant pipe 44a Branch part 45 First refrigerant pipe (first cooling part) 45a Gas reservoir 46 Second refrigerant pipe (second cooling part) 46a Gas reservoir 47 Regulating valve 47a First regulating valve 47b Second regulating valve 48 Compressor drive circuit 49 Fan 50 Heating mechanism 51 First heating part 52 Second heating part 55 First thermistor (first temperature sensor) 56 Second thermistor (second temperature sensor) 57 First holder 58 Second holder 60 Setting part 61 Display part 61a Battery state display part 61b External power supply connection display part 61c USB device power display part 61d Error display part 61e Right room temperature display part 61f Left room temperature display part 62 Right room temperature setting button 63 Left room temperature setting button 64 Mode switching button (room switching button) 65 Power button 66 USB device energization switching button 67 Execution button 70 Partition plate 71 Upper partition plate 72 Lower partition plate 73 Mating part 74 Heat insulating material 79 Accommodation object 80 Control circuit board 81 Microcomputer (operation control part) 82 Microcomputer (charging control part) 83 Control power supply 84 Rotation speed setting circuit 85 Shunt resistor 86a, 86b Battery voltage detection circuit 86c DC power supply voltage detection circuit 87 Regulating valve 88 Charging circuit 89 Shunt resistor 90 DC power supply 91 to 96 Hooking part 97 to 101 Groove part 102 Hooking part 147a First regulating valve 147b Second regulating valve 170 Partition plate 171 Upper partition plate 172 Lower Partition plate 173 Mating part 174 Heat insulating material 175 Hinge mechanism 200 Storage unit (additional part) 201, 201A Bag 202 Storage part 203 Grip part 204 to 206 Hook-and-loop fastener 208, 209 Cloth part 210 Button 221 Belt 222 Hook part (first engaging part) 224 Hook-and-loop fastener 226 Ring part (second engaging part) 230 Button 231 to 233 Accessory case 237 S-shaped hook 240, 241 Connection part 250 Storage unit (additional part) 251 Bag 253 Belt 254 Hook-and-loop fastener 300 Storage unit (additional part) 301 Bag 321 Belt 322 Hook part (first engaging part) 324 Hook-and-loop fastener 326 Ring part (second engaging part) 330 Button 400 Storage unit (additional part) 401 Bag 421 Belt 422 Hook part (first engaging part) 426 Ring part (second engaging part) 500 Storage unit (additional part) 501 Bag 521 Belt 522 Hook part 524 Hook-and-loop fastener 530 Button 621 Belt 622 Hook part 624 Hook-and-loop fastener 630 Button

Claims

1. An electric device comprising: a main body, having a first accommodation chamber and a second accommodation chamber; a lid, capable of being opened and closed with respect to the main body; a cooling mechanism that cools the first accommodation chamber and the second accommodation chamber; a control part that includes a microcomputer, which controls the cooling mechanism; and a setting part that includes a setting button, which allows a user to individually make set temperatures of the first accommodation chamber and the second accommodation chamber be set between a highest set temperature and a lowest set temperature respectively, wherein the control part is configured to set the set temperatures based on an operation of the setting part and control a set temperature difference between the set temperature of the first accommodation chamber and the set temperature of the second accommodation chamber to be within a predetermined value, wherein the user performs an operation to set the set temperature of the first accommodation chamber in the setting part, and when an operation is performed to cause the set temperature difference to exceed a predetermined value, the set temperature of the first accommodation chamber is not changed even if the operation is performed.
2. The electric device according to claim 1, further comprising: a display part that is controlled by the control part and individually displays the set temperatures of the first accommodation chamber and the second accommodation chamber, wherein the control part does not display the set

temperature at which the set temperature difference exceeds the predetermined value, even if an operation is performed on the setting part such that the set temperature difference exceeds the predetermined value.

3. The electric device according to claim 1, further comprising: a heating mechanism that heats the first accommodation chamber and the second accommodation chamber; wherein the control part is configured to control the heating mechanism.

4. The electric device according to claim 1, further comprising: a partition plate, configured to be detachable from the main body, and the first accommodation chamber and the second accommodation chamber are separated by the partition plate.

5. The electric device according to claim 4, wherein when the first accommodation chamber and the second accommodation chamber are used as one accommodation chamber without the partition plate, the setting part is configured to allow the user to set the set temperature of the one accommodation chamber between the highest set temperature and the lowest set temperature.

6. The electric device according to claim 5, further comprising: a display part, controlled by the control part to individually display the set temperatures of the first accommodation chamber and the second accommodation chamber, and to display the set temperature of the one accommodation chamber.

7. An electric device comprising: a main body, having a first accommodation chamber and a second accommodation chamber; a lid, capable of being opened and closed with respect to the main body; a cooling mechanism that cools the first accommodation chamber and the second accommodation chamber; a control part that includes a microcomputer, which controls the cooling mechanism; and a setting part that includes a setting button, which allows a user to individually make set temperatures of the first accommodation chamber and the second accommodation chamber be set between a highest set temperature and a lowest set temperature respectively, wherein the control part is configured to set the set temperatures based on an operation of the setting part and control a set temperature difference between the set temperature of the first accommodation chamber and the set temperature of the second accommodation chamber to be within a predetermined value, wherein the user performs an operation to set the set temperature of the first accommodation chamber in the setting part, and when an operation is performed to cause the set temperature difference to exceed a predetermined value, along with the operation, the set temperature of the first accommodation chamber is set, and the set temperature of the second accommodation chamber is automatically changed.

8. The electric device according to claim 7, wherein further comprising: a display part that is controlled by the control part and individually displays the set temperatures of the first accommodation chamber and the second accommodation chamber, wherein the control part is configured to be operated by the user to set the set temperature of the first accommodation chamber to the setting part, when an operation is performed such that the set temperature difference exceeds the predetermined value, in accordance with the operation, a set temperature display of the first accommodation chamber is switched, and a set temperature display of the second accommodation chamber is automatically switched.

9. The electric device according to claim 7, further comprising: a heating mechanism that heats the first accommodation chamber and the second accommodation chamber; wherein the control part is configured to control the heating mechanism.

10. The electric device according to claim 7, further comprising: a partition plate, configured to be detachable from the main body, and the first accommodation chamber and the second accommodation chamber are separated by the partition plate.

11. The electric device according to claim 7, further comprising: a display part that is controlled by the control part and individually displays the set temperatures of the first accommodation chamber and the second accommodation chamber, wherein the control part does not display the set temperature at which the set temperature difference exceeds the predetermined value, even if an

operation is performed on the setting part such that the set temperature difference exceeds the predetermined value.

12. The electric device according to claim 10, wherein when the first accommodation chamber and the second accommodation chamber are used as one accommodation chamber without the partition plate, the setting part is configured to allow the user to set the set temperature of the one accommodation chamber between the highest set temperature and the lowest set temperature.

13. The electric device according to claim 12, wherein a display part, controlled by the control part to individually display the set temperatures of the first accommodation chamber and the second accommodation chamber, and to display the set temperature of the one accommodation chamber.
